C Programming Lab Manual

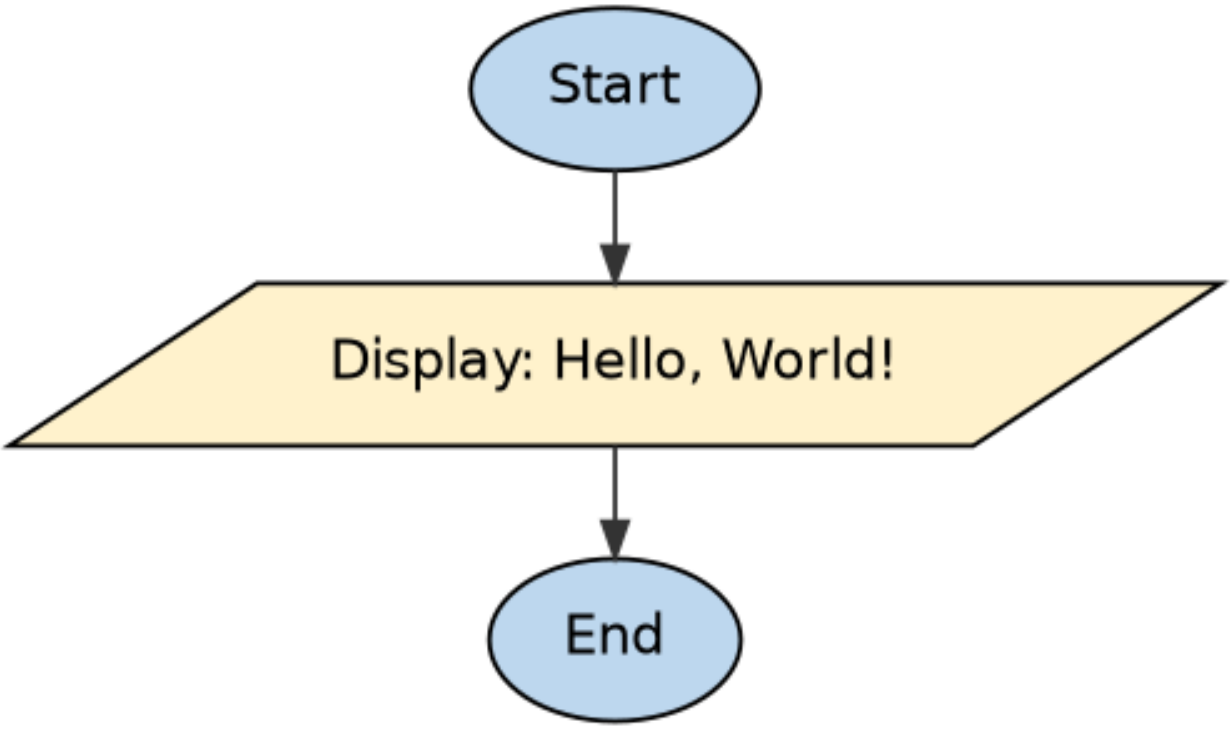
Solutions: Lab 01 to Lab 20 (40 Programs)

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Lab 01 – Basic C Programming

Question 1.1: Write a C program to print "Hello, World!".

Flowchart



Program

```
#include <stdio.h>

int main()
{
    printf("Hello, World!");
    return 0;
}
```

Input

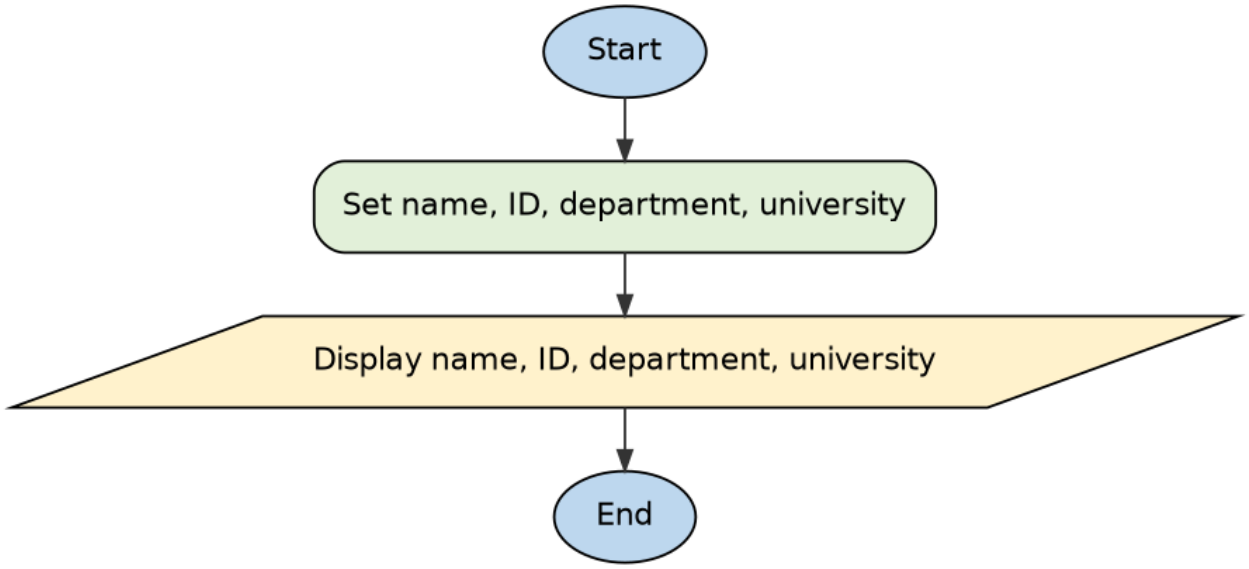
```
No input required.
```

Output

```
Hello, World!
```

Question 1.2: Write a C program to display your name, ID, department, and university name.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    char name[] = "Your Name";
    int id = 12345;               /* replace with your student ID */
    char dept[] = "Your Department";
    char university[] = "Your University";

    printf("Name       : %s\n", name);
    printf("ID         : %d\n", id);
    printf("Department : %s\n", dept);
    printf("University : %s\n", university);
    return 0;
}
```

Input

```
No input required (values are fixed in the program; edit them to your own details).
```

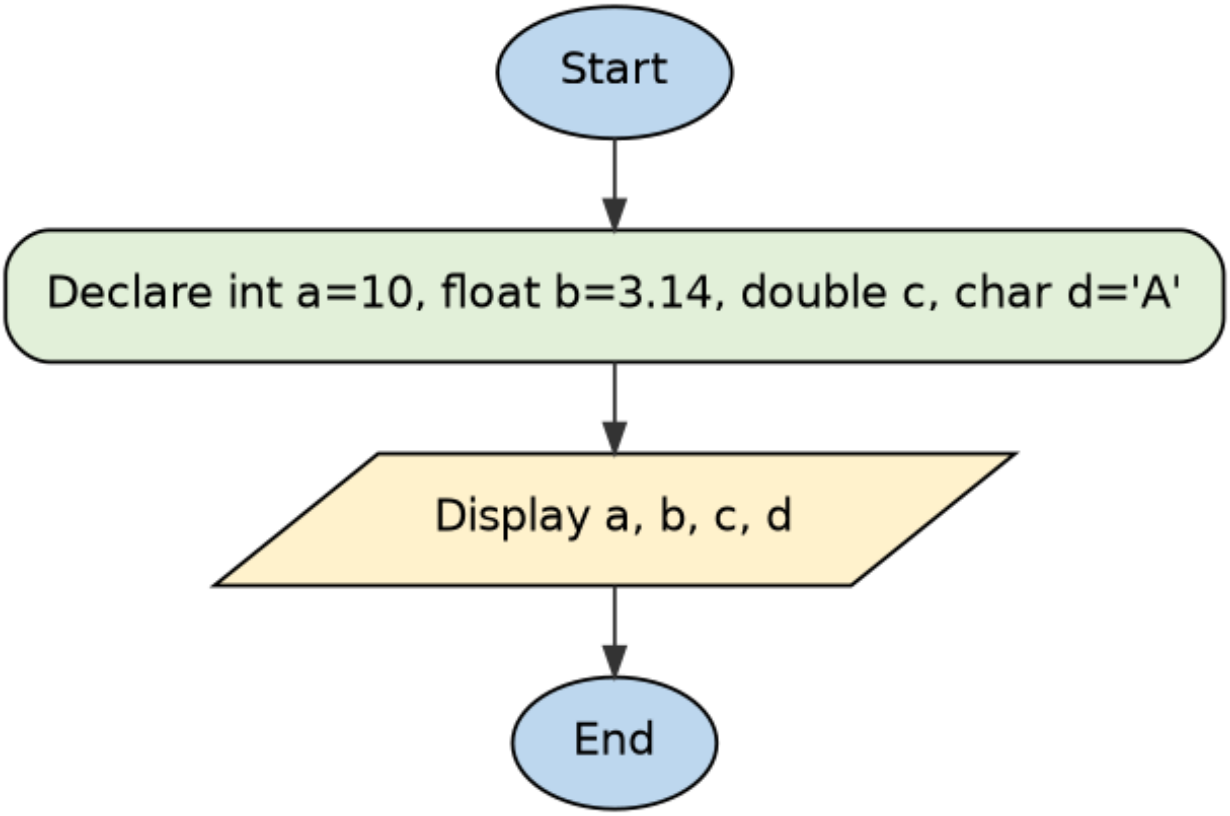
Output

```
Name       : Your Name
ID         : 12345
Department : Your Department
University : Your University
```

Lab 02 – Variables, Constants and Data Types

Question 2.1: Write a C program to declare and display variables of different data types.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a = 10;
    float b = 3.14f;
    double c = 2.718281828;
    char d = 'A';

    printf("int    a = %d\n", a);
    printf("float  b = %f\n", b);
    printf("double c = %lf\n", c);
    printf("char   d = %c\n", d);
    return 0;
}
```

Input

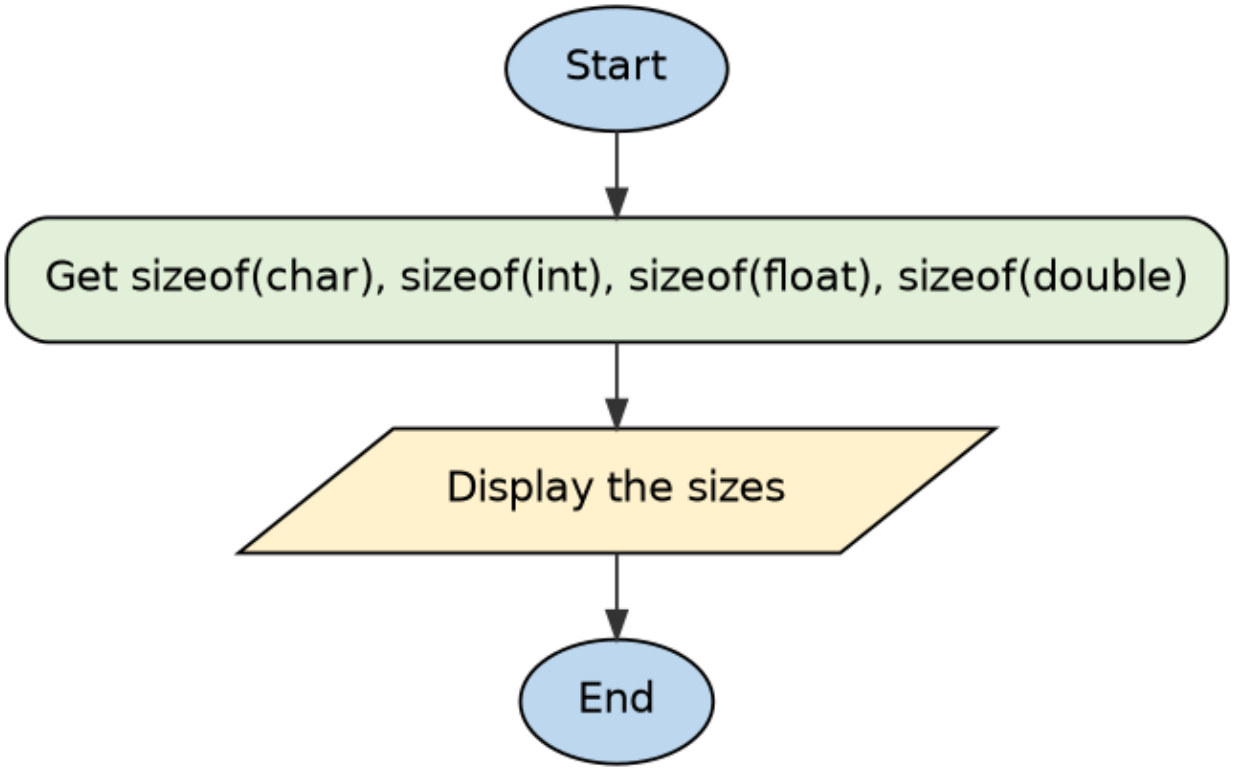
```
No input required.
```

Output

```
int    a = 10
float  b = 3.140000
double c = 2.718282
char   d = A
```

Question 2.2: Write a C program to display the size of different data types using sizeof().

Flowchart



Program

```
#include <stdio.h>

int main()
{
    printf("Size of char   : %zu byte(s)\n", sizeof(char));
    printf("Size of int    : %zu byte(s)\n", sizeof(int));
    printf("Size of float  : %zu byte(s)\n", sizeof(float));
    printf("Size of double : %zu byte(s)\n", sizeof(double));
    return 0;
}
```

Input

```
No input required.
```

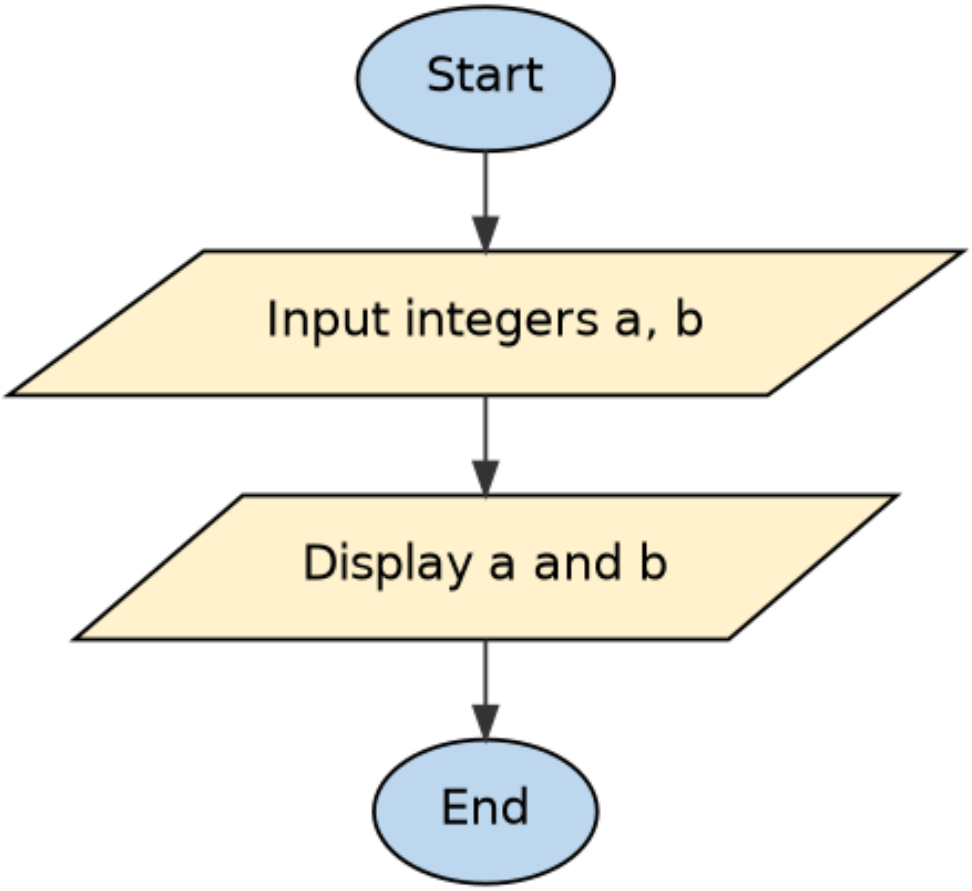
Output

```
Size of char   : 1 byte(s)
Size of int    : 4 byte(s)
Size of float  : 4 byte(s)
Size of double : 8 byte(s)
(Values may vary by compiler/system.)
```

Lab 03 – Input and Output

Question 3.1: Write a C program to take two integers as input and display them.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a, b;

    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);

    printf("First integer  : %d\n", a);
    printf("Second integer : %d\n", b);
    return 0;
}
```

Input

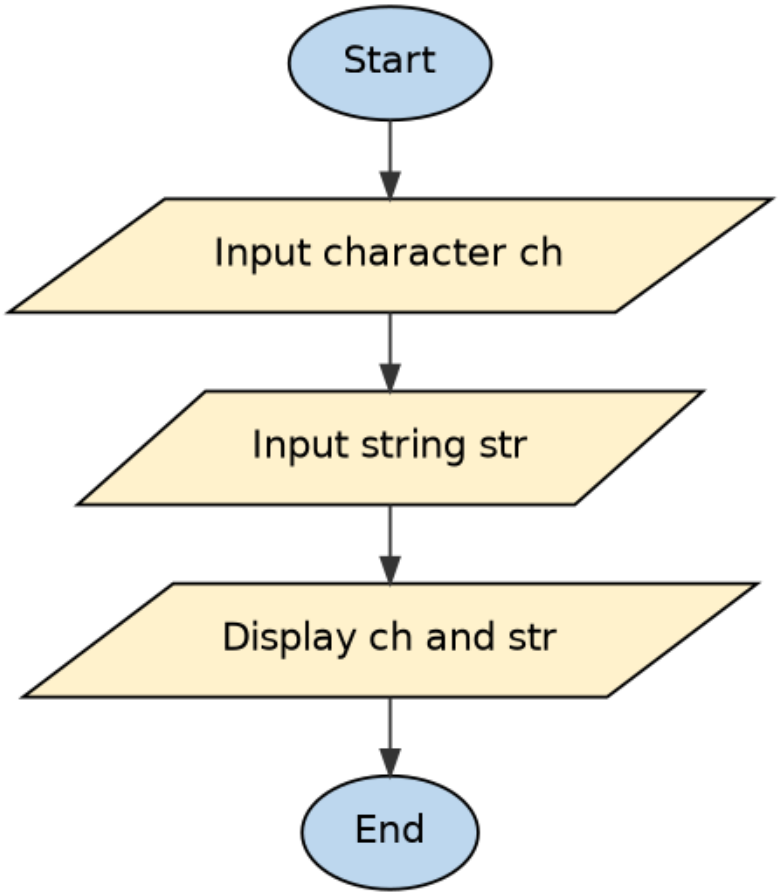
```
Enter two integers: 15 27
```

Output

```
First integer  : 15
Second integer : 27
```

Question 3.2: Write a C program to take a character and a string as input and display them.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    char ch;
    char str[100];

    printf("Enter a character: ");
    scanf(" %c", &ch);          /* leading space skips leftover newline */

    printf("Enter a string: ");
    scanf("%s", str);

    printf("Character : %c\n", ch);
    printf("String    : %s\n", str);
    return 0;
}
```

Input

```
Enter a character: K
Enter a string: Hello
```

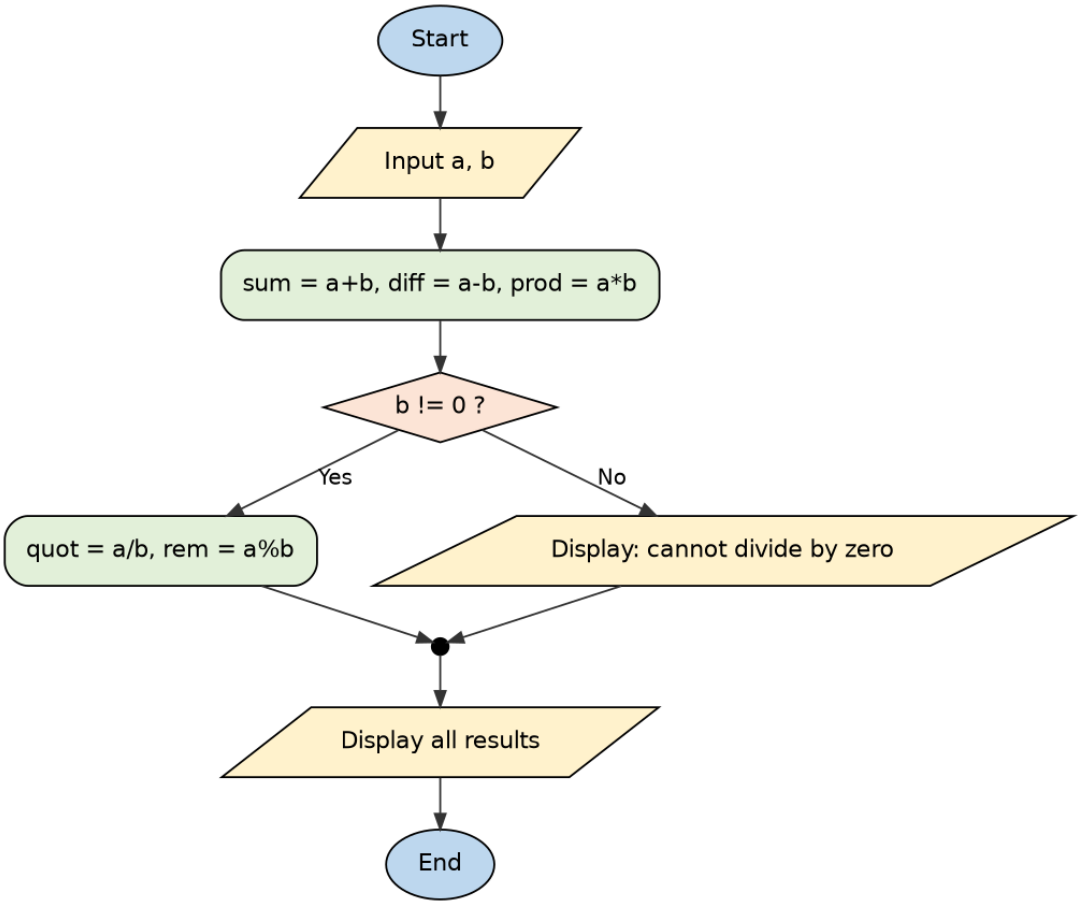
Output

```
Character : K
String    : Hello
```

Lab 04 – Arithmetic Operators

Question 4.1: Write a C program to perform addition, subtraction, multiplication, division, and modulus of two numbers.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a, b;

    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);

    printf("Addition       : %d + %d = %d\n", a, b, a + b);
    printf("Subtraction    : %d - %d = %d\n", a, b, a - b);
    printf("Multiplication : %d * %d = %d\n", a, b, a * b);
    if (b != 0) {
        printf("Division       : %d / %d = %d\n", a, b, a / b);
        printf("Modulus        : %d %% %d = %d\n", a, b, a % b);
    } else {
        printf("Division and modulus not possible (divisor is zero).\n");
    }
    return 0;
}
```

Input

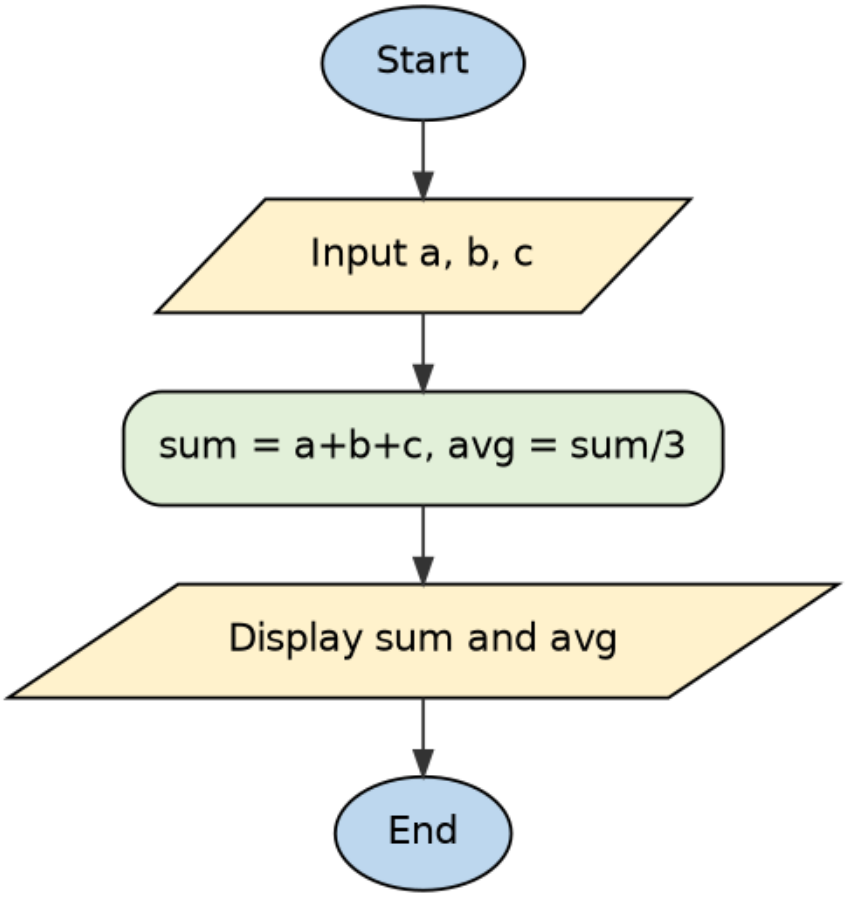
```
Enter two integers: 17 5
```

Output

```
Addition       : 17 + 5 = 22
Subtraction    : 17 - 5 = 12
Multiplication : 17 * 5 = 85
Division       : 17 / 5 = 3
Modulus        : 17 % 5 = 2
```

Question 4.2: Write a C program to calculate the sum and average of three numbers.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    float a, b, c, sum, avg;

    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);

    sum = a + b + c;
    avg = sum / 3;

    printf("Sum     = %.2f\n", sum);
    printf("Average = %.2f\n", avg);
    return 0;
}
```

Input

```
Enter three numbers: 10 20 30
```

Output

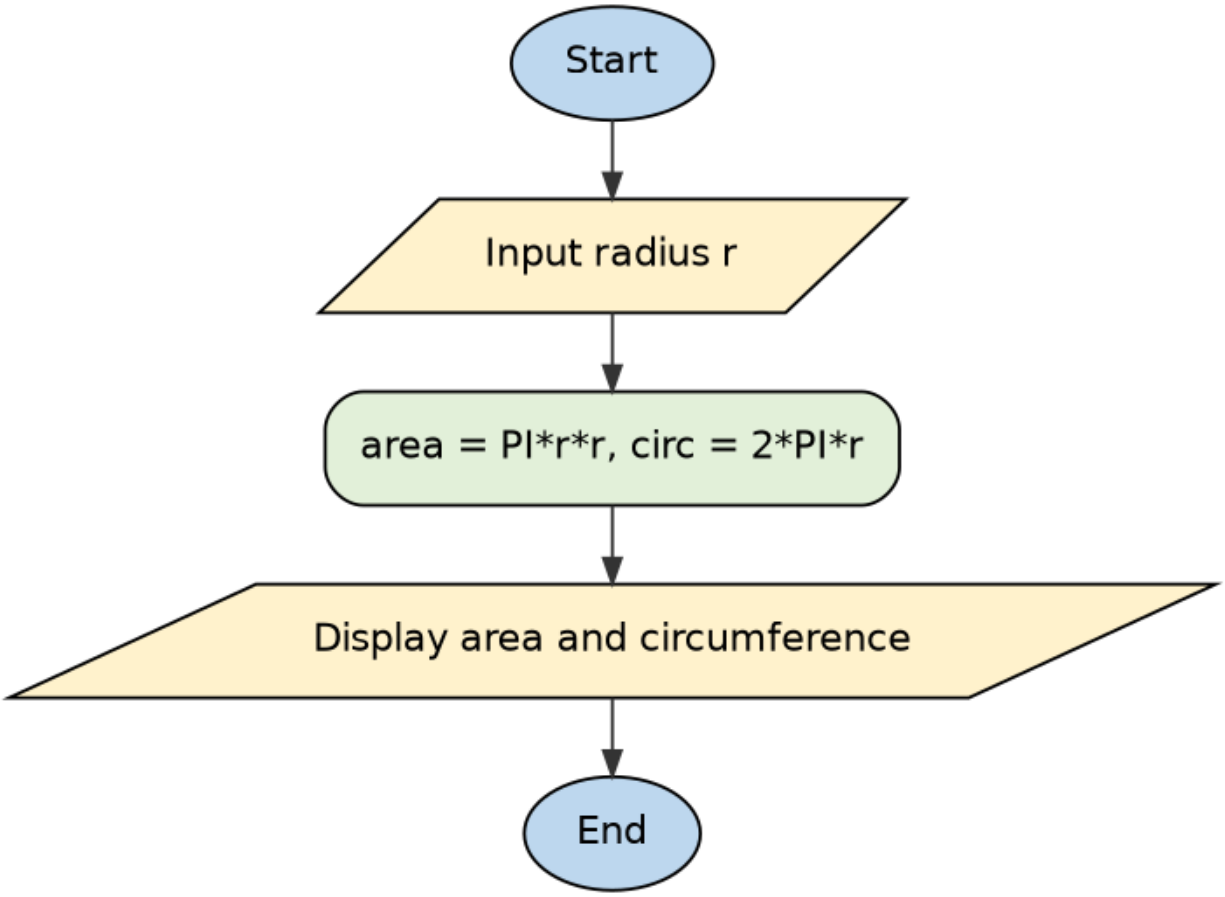
```
Sum     = 60.00
```

```
Average = 20.00
```

Lab 05 – More Arithmetic Problems

Question 5.1: Write a C program to calculate the area and circumference of a circle.

Flowchart



Program

```
#include <stdio.h>

#define PI 3.14159

int main()
{
    float radius, area, circumference;

    printf("Enter the radius of the circle: ");
    scanf("%f", &radius);

    area = PI * radius * radius;
    circumference = 2 * PI * radius;

    printf("Area          = %.2f\n", area);
    printf("Circumference = %.2f\n", circumference);
    return 0;
}
```

Input

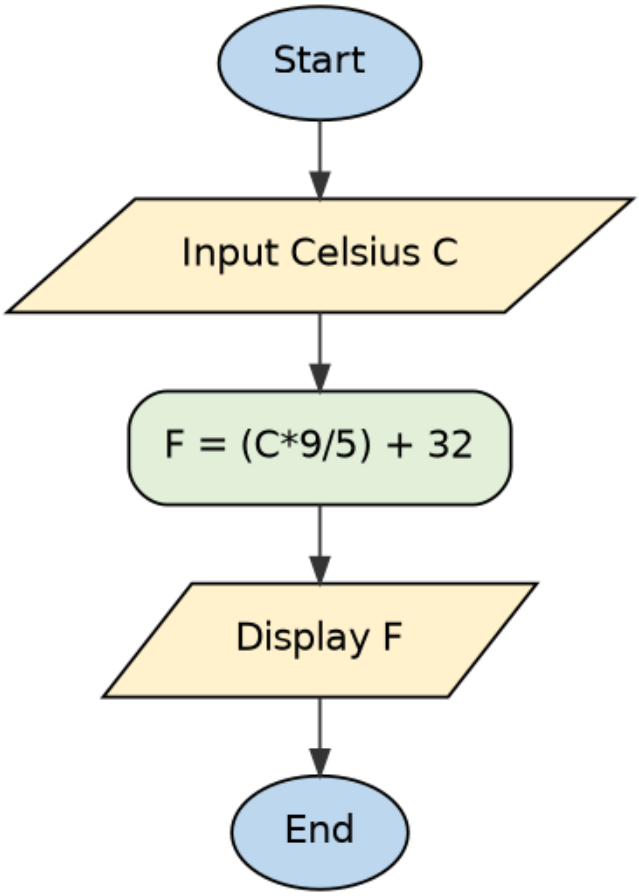
```
Enter the radius of the circle: 7
```

Output

```
Area          = 153.94
Circumference = 43.98
```

Question 5.2: Write a C program to convert Celsius to Fahrenheit.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    float celsius, fahrenheit;

    printf("Enter temperature in Celsius: ");
    scanf("%f", &celsius);

    fahrenheit = (celsius * 9 / 5) + 32;

    printf("%.2f Celsius = %.2f Fahrenheit\n", celsius, fahrenheit);
    return 0;
}
```

Input

```
Enter temperature in Celsius: 37
```

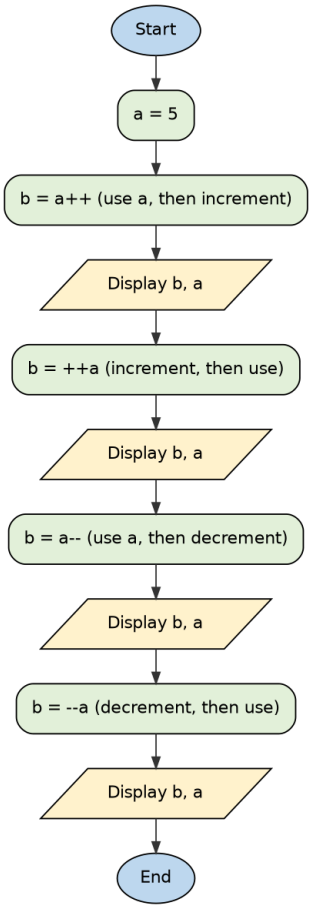
Output

```
37.00 Celsius = 98.60 Fahrenheit
```

Lab 06 – Increment, Decrement and Assignment Operators

Question 6.1: Write a C program to demonstrate the use of increment and decrement operators.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a = 5, b;

    printf("Initial a = %d\n", a);

    b = a++;                    /* post-increment: use then increment */
    printf("b = a++  -> b = %d, a = %d\n", b, a);

    b = ++a;                    /* pre-increment: increment then use */
    printf("b = ++a  -> b = %d, a = %d\n", b, a);

    b = a--;                    /* post-decrement */
    printf("b = a--  -> b = %d, a = %d\n", b, a);

    b = --a;                    /* pre-decrement */
    printf("b = --a  -> b = %d, a = %d\n", b, a);
    return 0;
}
```

Input

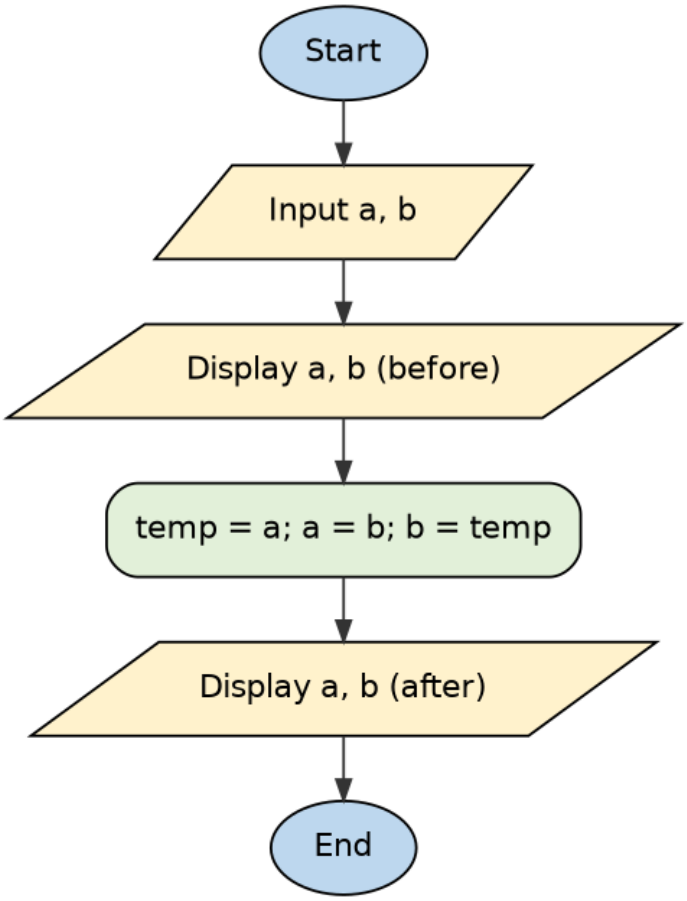
```
No input required.
```

Output

```
Initial a = 5
b = a++  -> b = 5, a = 6
b = ++a  -> b = 7, a = 7
b = a--  -> b = 7, a = 6
b = --a  -> b = 5, a = 5
```

Question 6.2: Write a C program to swap two numbers using a third variable.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a, b, temp;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Before swap: a = %d, b = %d\n", a, b);

    temp = a;
    a = b;
    b = temp;

    printf("After swap : a = %d, b = %d\n", a, b);
    return 0;
}
```

Input

```
Enter two numbers: 4 9
```

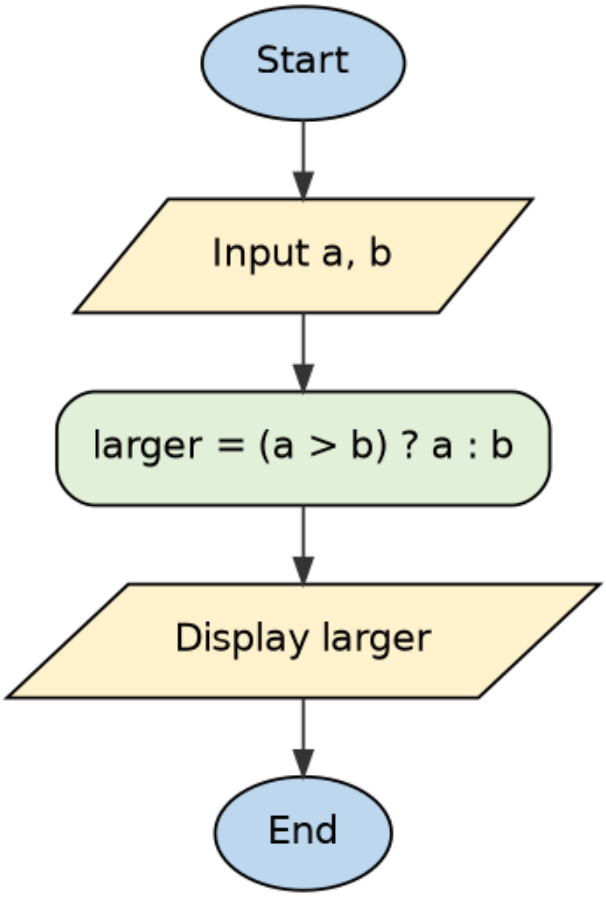
Output

```
Before swap: a = 4, b = 9
After swap : a = 9, b = 4
```

Lab 07 – Conditional Operator

Question 7.1: Write a C program to find the larger of two numbers using the conditional operator.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a, b, larger;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    larger = (a > b) ? a : b;

    printf("Larger number = %d\n", larger);
    return 0;
}
```

Input

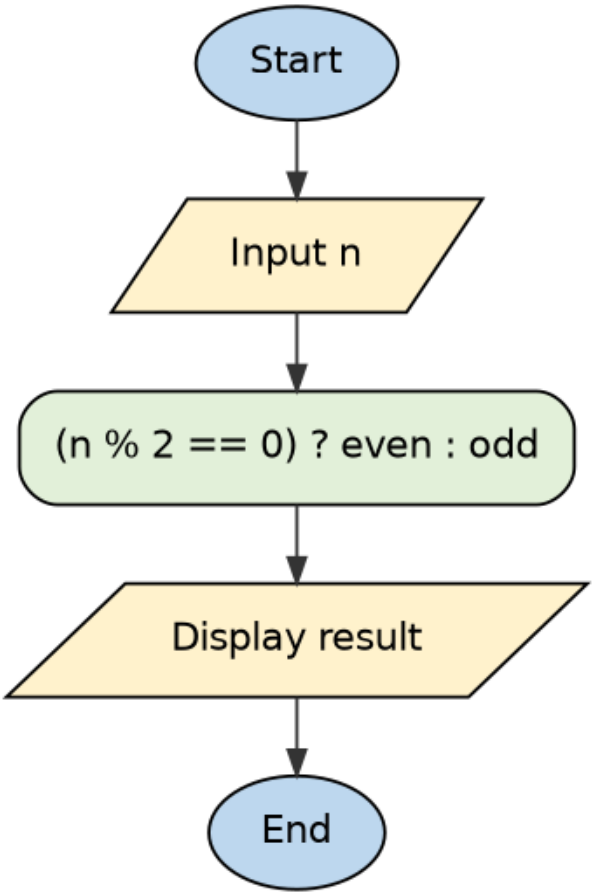
```
Enter two numbers: 45 82
```

Output

```
Larger number = 82
```

Question 7.2: Write a C program to check whether a number is even or odd using the conditional operator.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter an integer: ");
    scanf("%d", &n);

    (n % 2 == 0) ? printf("%d is even.\n", n) : printf("%d is odd.\n", n);
    return 0;
}
```

Input

```
Enter an integer: 13
```

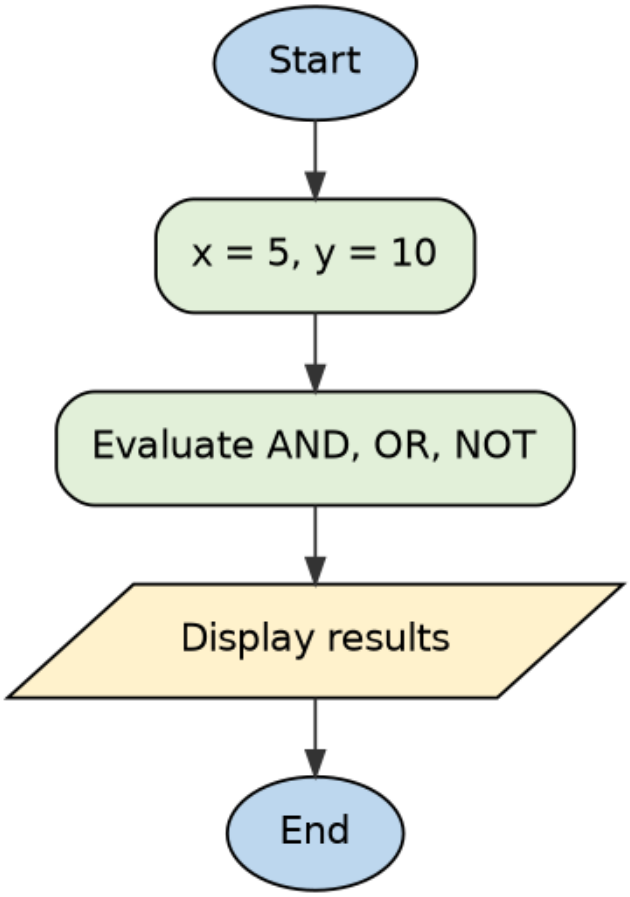
Output

```
13 is odd.
```

Lab 08 – Relational and Logical Operators

Question 8.1: Write a C program to demonstrate the use of logical AND, OR, and NOT operators.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int x = 5, y = 10;

    printf("x = %d, y = %d\n", x, y);
    printf("(x > 0 && y > 0) : %d\n", x > 0 && y > 0);   /* AND */
    printf("(x > 7 || y > 7) : %d\n", x > 7 || y > 7);   /* OR  */
    printf("!(x == y)        : %d\n", !(x == y));        /* NOT */
    return 0;
}
```

Input

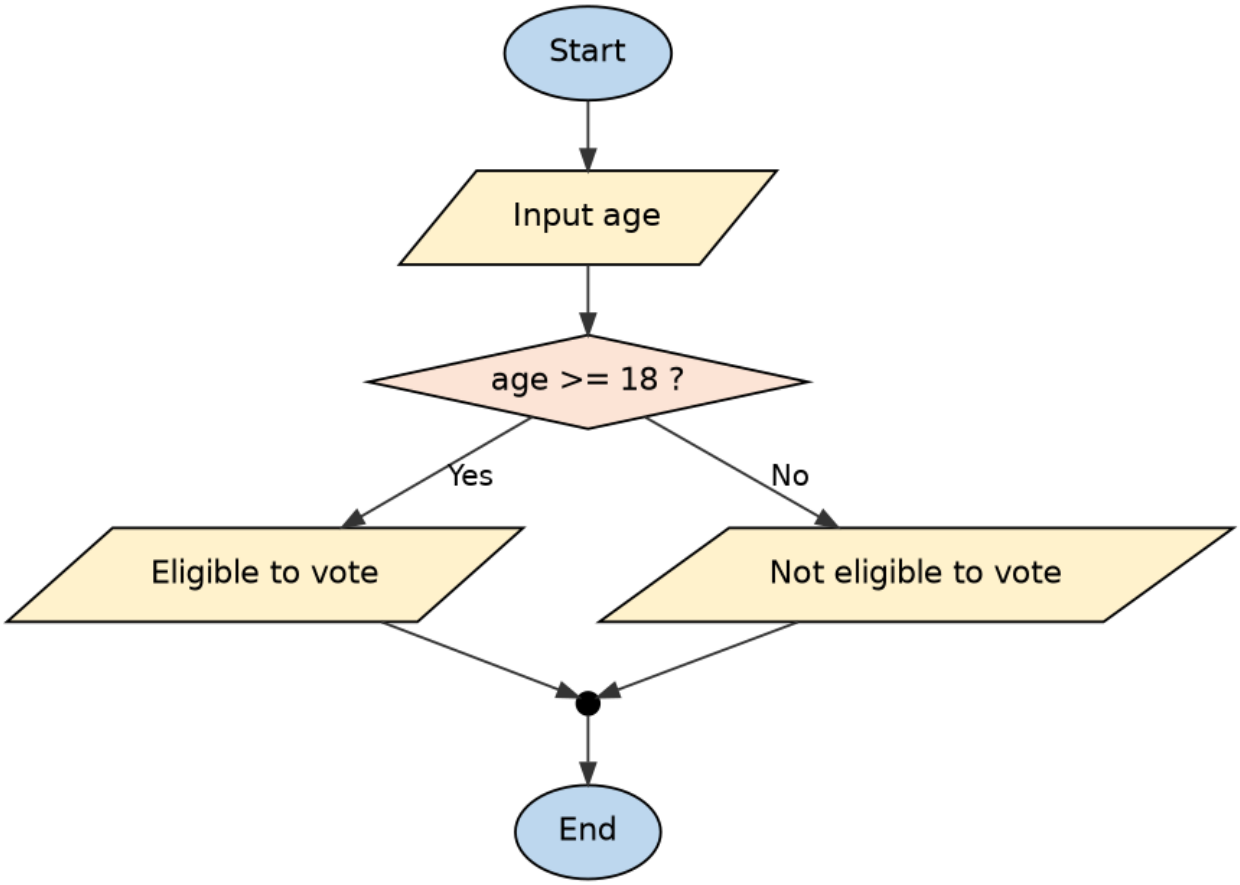
```
No input required.
```

Output

```
x = 5, y = 10
(x > 0 && y > 0) : 1
(x > 7 || y > 7) : 1
!(x == y)        : 1
(1 = true, 0 = false)
```

Question 8.2: Write a C program to check whether a person is eligible to vote.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int age;

    printf("Enter your age: ");
    scanf("%d", &age);

    if (age >= 18)
        printf("You are eligible to vote.\n");
    else
        printf("You are not eligible to vote.\n");
    return 0;
}
```

Input

```
Enter your age: 20
```

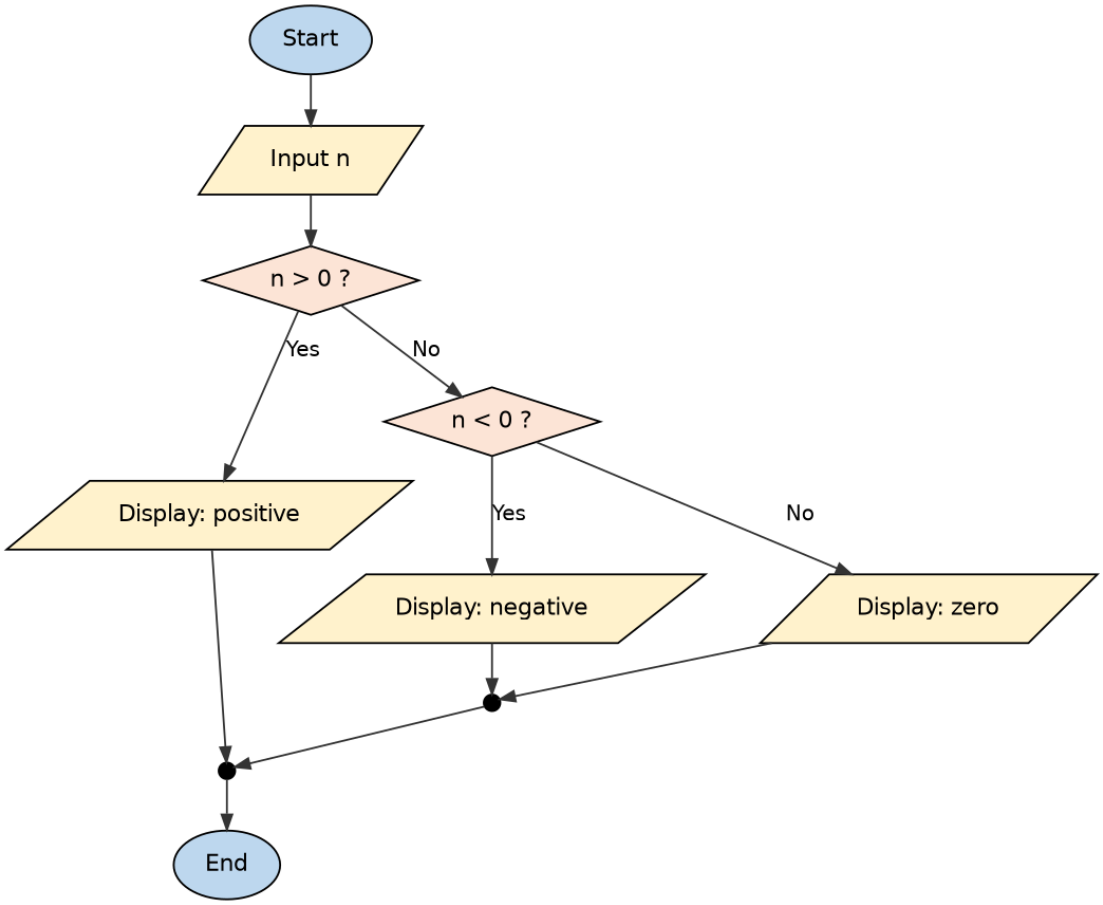
Output

```
You are eligible to vote.
```

Lab 09 – if Statement

Question 9.1: Write a C program to check whether a number is positive, negative, or zero.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter a number: ");
    scanf("%d", &n);

    if (n > 0)
        printf("%d is positive.\n", n);
    if (n < 0)
        printf("%d is negative.\n", n);
    if (n == 0)
        printf("The number is zero.\n");
    return 0;
}
```

Input

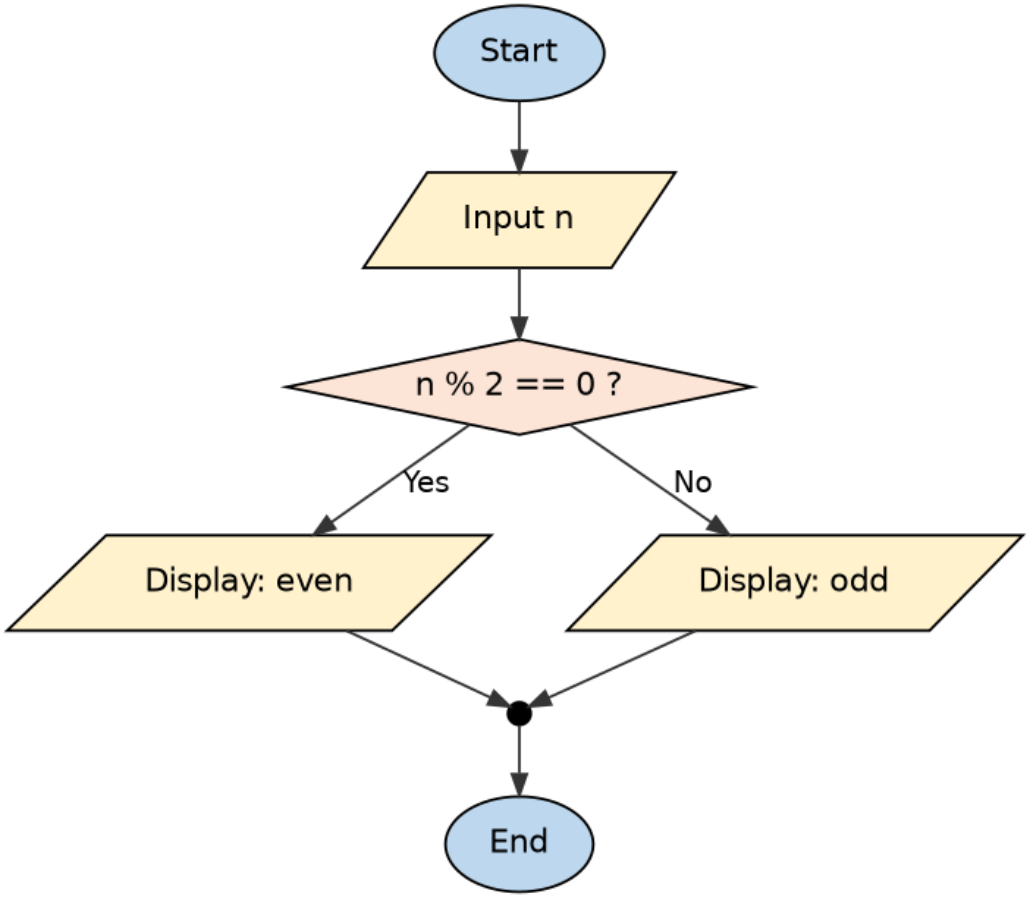
```
Enter a number: -8
```

Output

```
-8 is negative.
```

Question 9.2: Write a C program to check whether a number is even or odd.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter an integer: ");
    scanf("%d", &n);

    if (n % 2 == 0)
        printf("%d is even.\n", n);
    if (n % 2 != 0)
        printf("%d is odd.\n", n);
    return 0;
}
```

Input

```
Enter an integer: 28
```

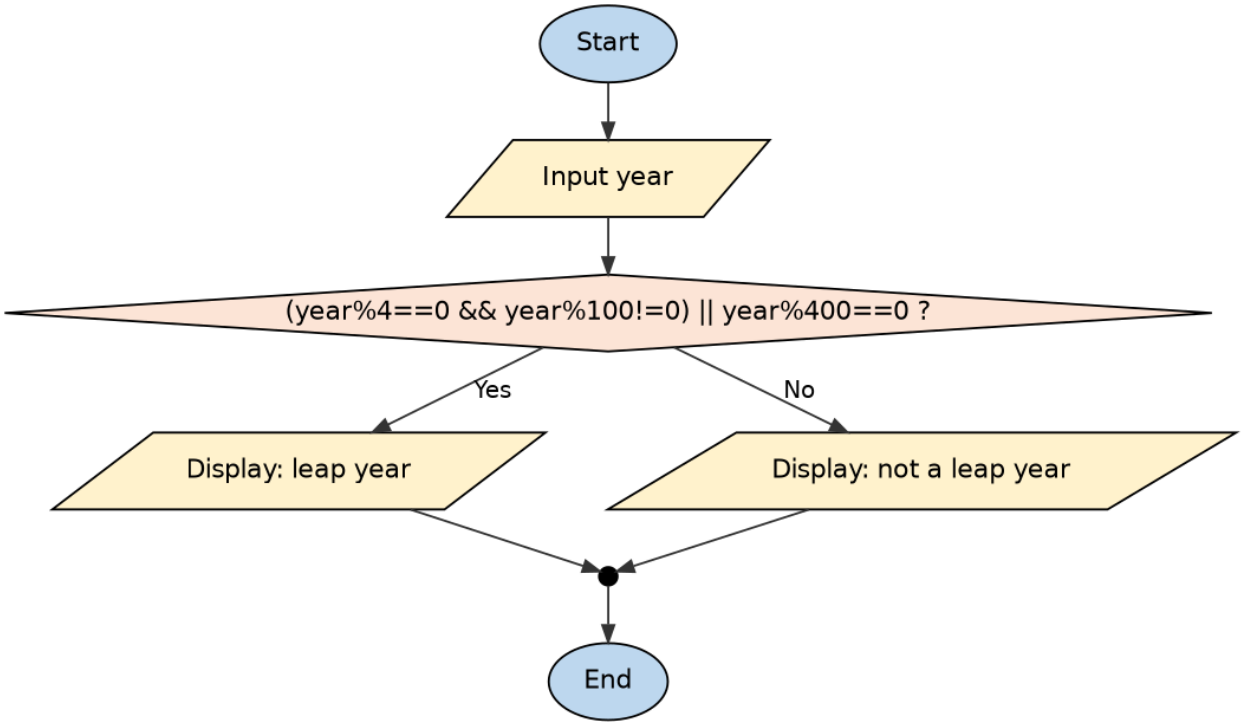
Output

```
28 is even.
```

Lab 10 – if-else Statement

Question 10.1: Write a C program to check whether a year is a leap year or not.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
        printf("%d is a leap year.\n", year);
    else
        printf("%d is not a leap year.\n", year);
    return 0;
}
```

Input

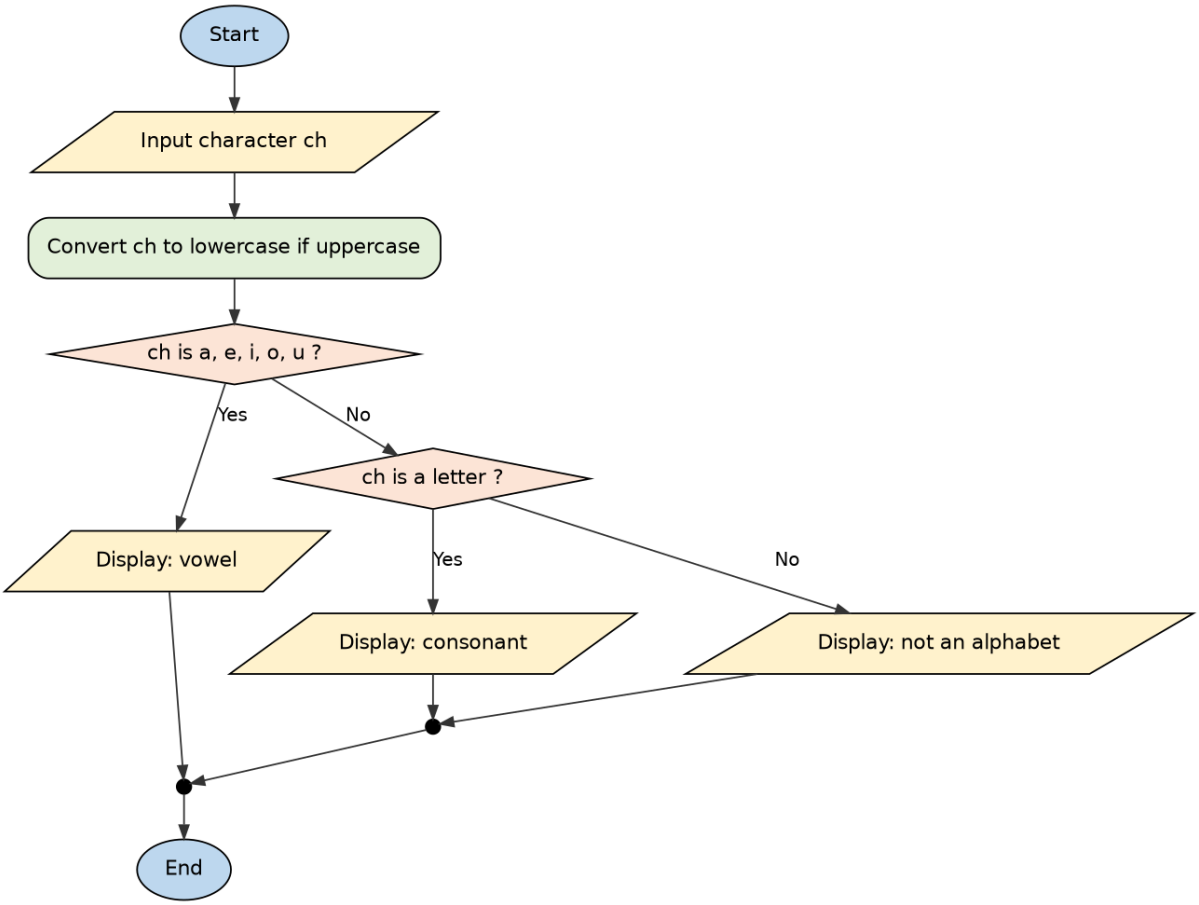
```
Enter a year: 2024
```

Output

```
2024 is a leap year.
```

Question 10.2: Write a C program to check whether a character is a vowel or consonant.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    char ch;

    printf("Enter a character: ");
    scanf(" %c", &ch);

    if (ch >= 'A' && ch <= 'Z')
        ch = ch + 32;                /* convert to lowercase */

    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
        printf("'%c' is a vowel.\n", ch);
    else if (ch >= 'a' && ch <= 'z')
        printf("'%c' is a consonant.\n", ch);
    else
        printf("Not an alphabet character.\n");
    return 0;
}
```

Input

```
Enter a character: E
```

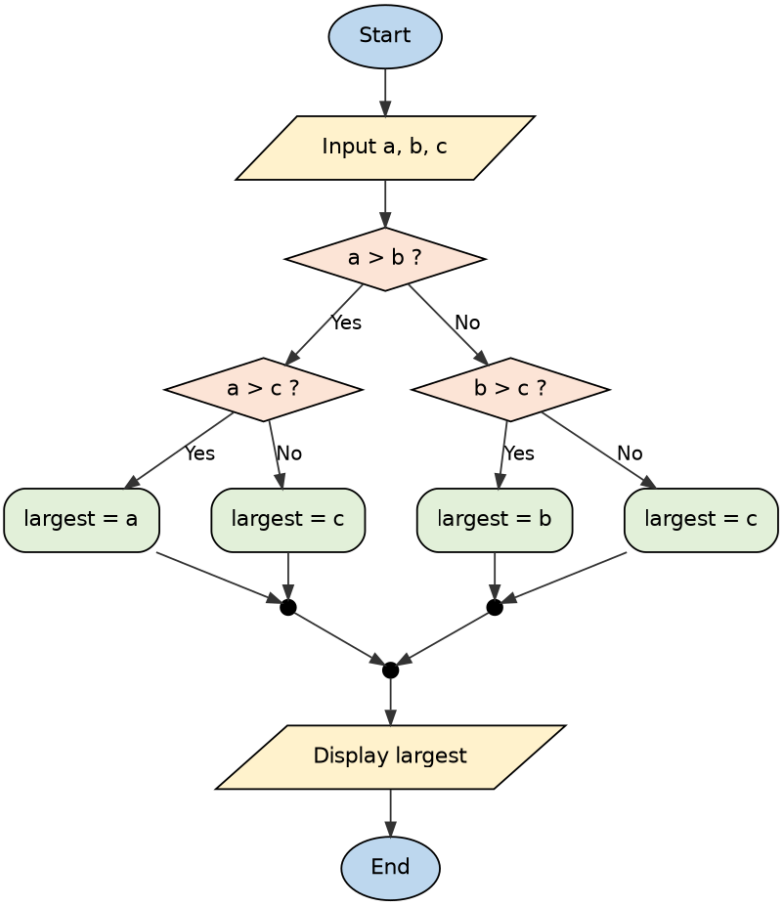
Output

```
'e' is a vowel.
```

Lab 11 – Nested if

Question 11.1: Write a C program to find the largest among three numbers using nested if.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a, b, c;

    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);

    if (a > b) {
        if (a > c)
            printf("Largest = %d\n", a);
        else
            printf("Largest = %d\n", c);
    } else {
        if (b > c)
            printf("Largest = %d\n", b);
        else
            printf("Largest = %d\n", c);
    }
    return 0;
}
```

Input

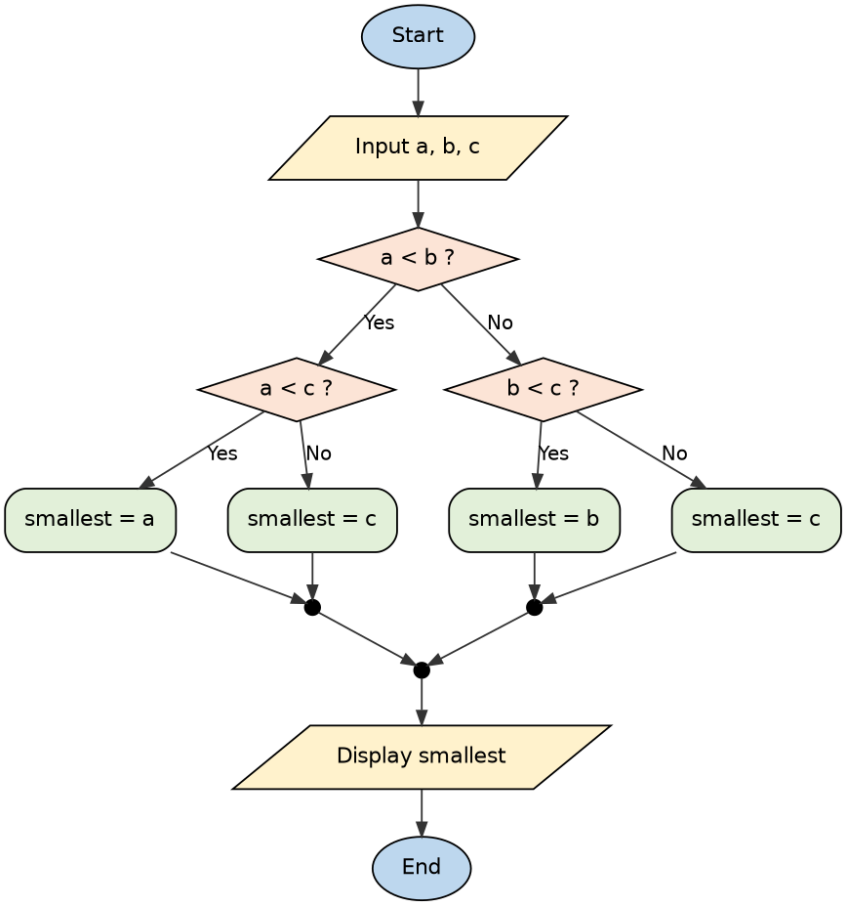
```
Enter three numbers: 12 45 30
```

Output

```
Largest = 45
```

Question 11.2: Write a C program to find the smallest among three numbers using nested if.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a, b, c;

    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);

    if (a < b) {
        if (a < c)
            printf("Smallest = %d\n", a);
        else
            printf("Smallest = %d\n", c);
    } else {
        if (b < c)
            printf("Smallest = %d\n", b);
        else
            printf("Smallest = %d\n", c);
    }
    return 0;
}
```

Input

```
Enter three numbers: 12 45 3
```

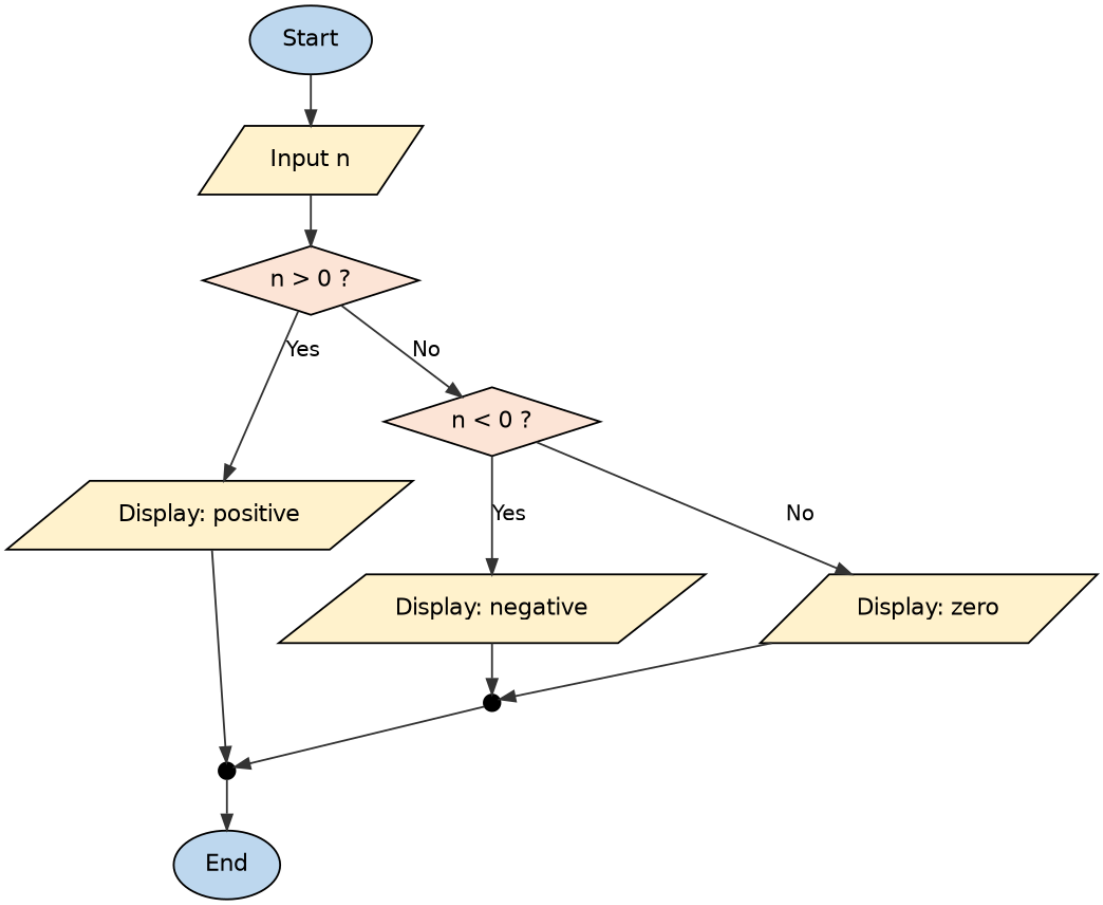
Output

```
Smallest = 3
```

Lab 12 – else-if Ladder

Question 12.1: Write a C program to determine whether a number is positive, negative, or zero using an else-if ladder.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter a number: ");
    scanf("%d", &n);

    if (n > 0)
        printf("%d is positive.\n", n);
    else if (n < 0)
        printf("%d is negative.\n", n);
    else
        printf("The number is zero.\n");
    return 0;
}
```

Input

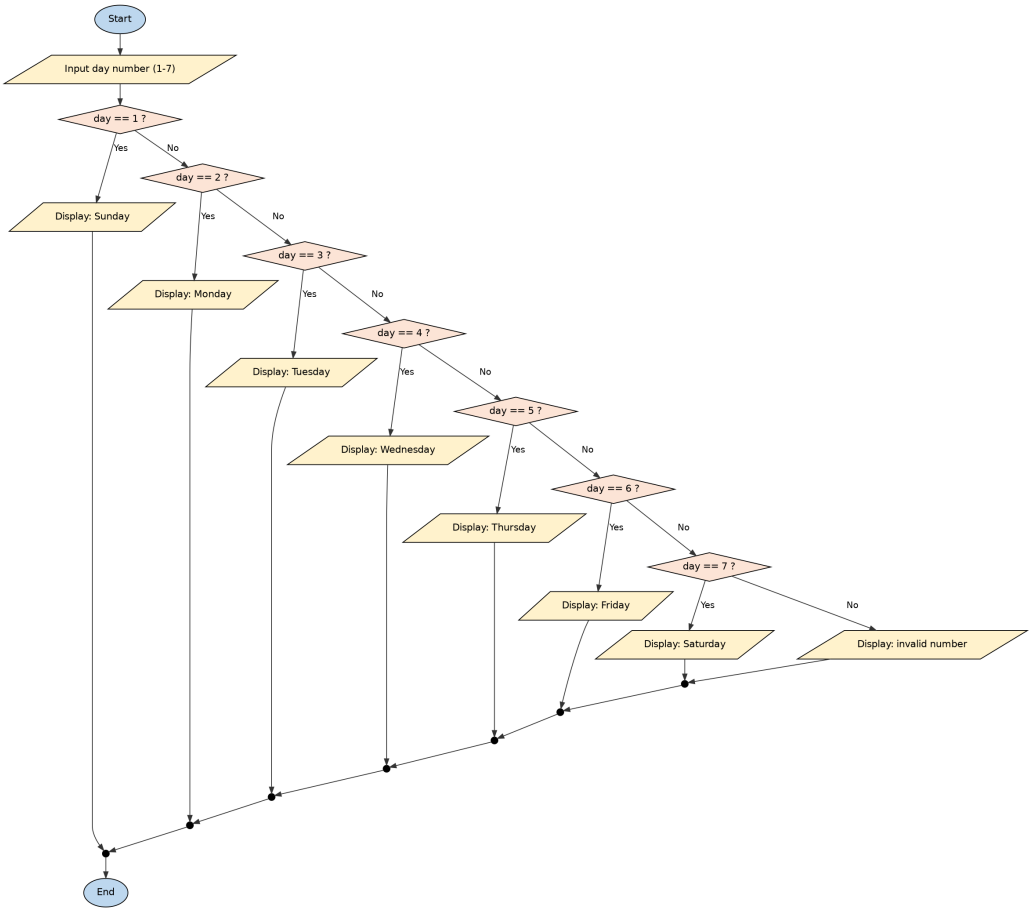
```
Enter a number: 0
```

Output

```
The number is zero.
```

Question 12.2: Write a C program to display the day of the week based on a number from 1 to 7.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int day;

    printf("Enter a number (1-7): ");
    scanf("%d", &day);

    if (day == 1)
        printf("Sunday\n");
    else if (day == 2)
        printf("Monday\n");
    else if (day == 3)
        printf("Tuesday\n");
    else if (day == 4)
        printf("Wednesday\n");
    else if (day == 5)
        printf("Thursday\n");
    else if (day == 6)
        printf("Friday\n");
    else if (day == 7)
        printf("Saturday\n");
    else
        printf("Invalid input! Enter a number from 1 to 7.\n");
    return 0;
}
```

Input

```
Enter a number (1-7): 5
```

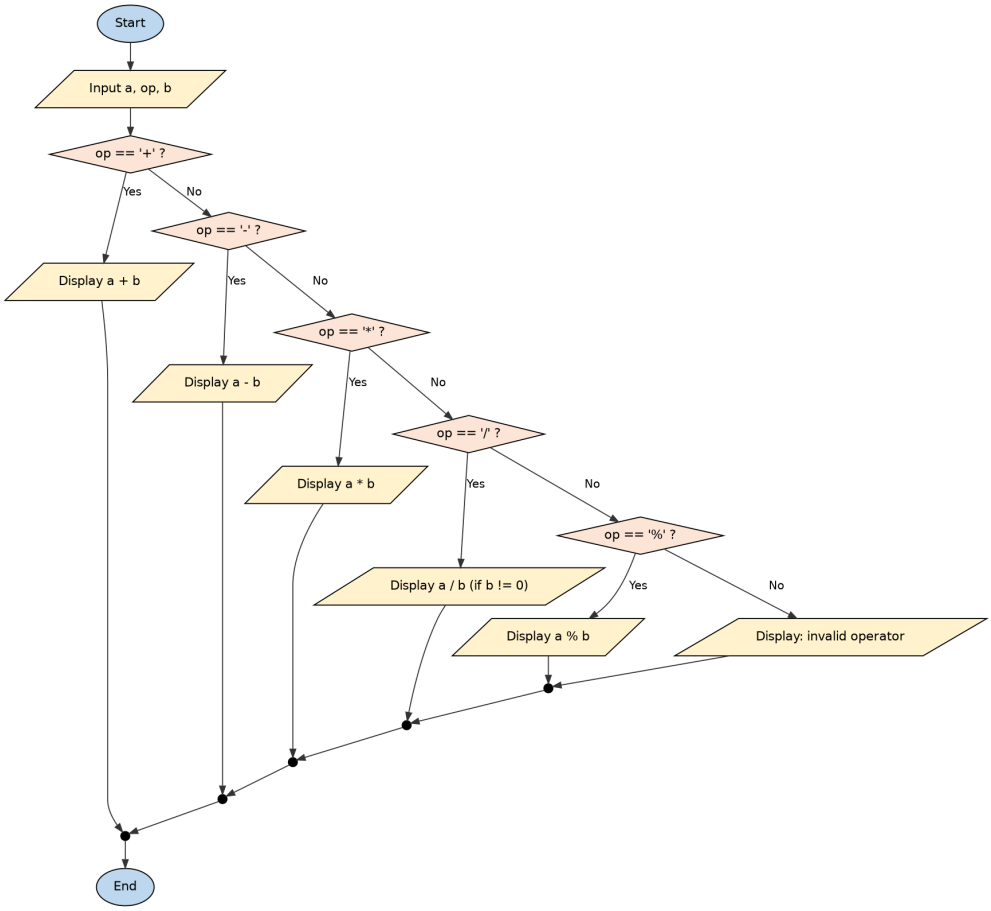
Output

```
Thursday
```

Lab 13 – switch Statement

Question 13.1: Write a C program to perform basic arithmetic operations using switch.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a, b, result;
    char op;

    printf("Enter expression (e.g. 8 + 4): ");
    scanf("%d %c %d", &a, &op, &b);

    switch (op) {
        case '+': result = a + b; printf("Result = %d\n", result); break;
        case '-': result = a - b; printf("Result = %d\n", result); break;
        case '*': result = a * b; printf("Result = %d\n", result); break;
        case '/':
            if (b != 0) printf("Result = %d\n", a / b);
            else        printf("Error: division by zero.\n");
            break;
        case '%': result = a % b; printf("Result = %d\n", result); break;
        default:  printf("Invalid operator.\n");
    }
    return 0;
}
```

Input

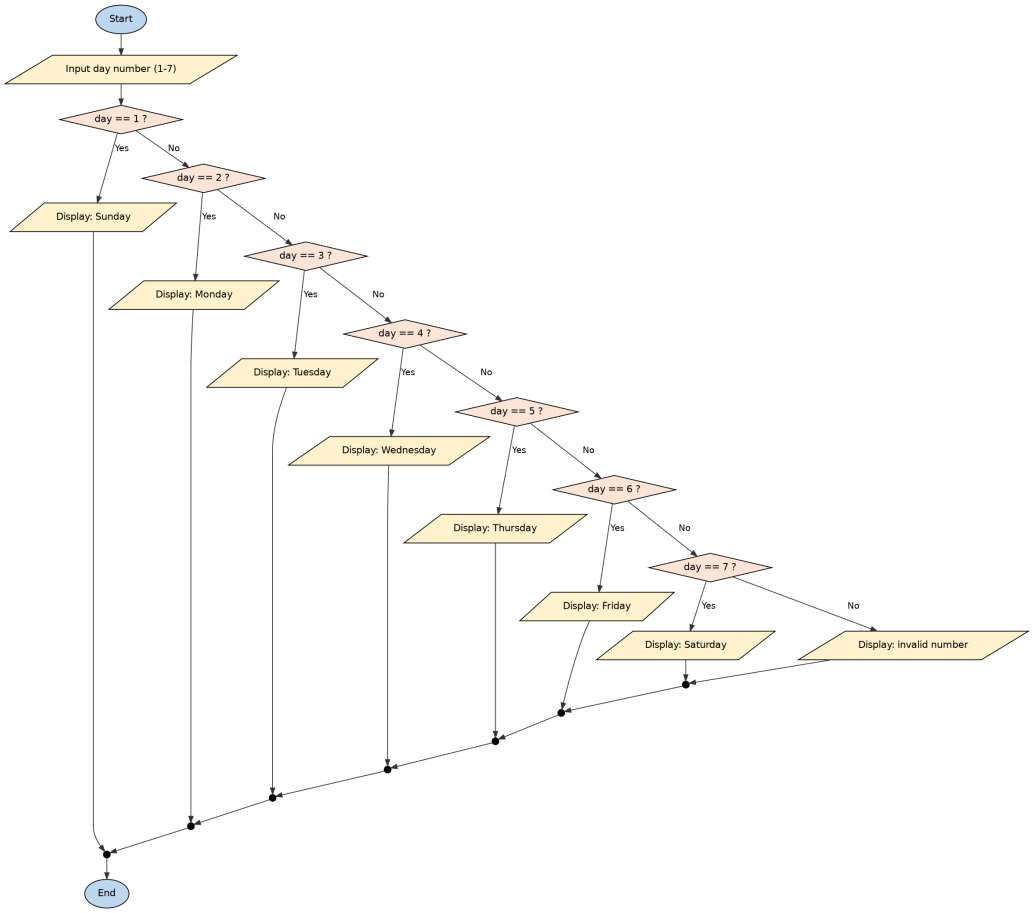
```
Enter expression (e.g. 8 + 4): 8 * 4
```

Output

```
Result = 32
```

Question 13.2: Write a C program to display the day of the week using switch.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int day;

    printf("Enter a number (1-7): ");
    scanf("%d", &day);

    switch (day) {
        case 1: printf("Sunday\n");    break;
        case 2: printf("Monday\n");    break;
        case 3: printf("Tuesday\n");   break;
        case 4: printf("Wednesday\n"); break;
        case 5: printf("Thursday\n");  break;
        case 6: printf("Friday\n");    break;
        case 7: printf("Saturday\n");  break;
        default: printf("Invalid day number.\n");
    }
    return 0;
}
```

Input

```
Enter a number (1-7): 2
```

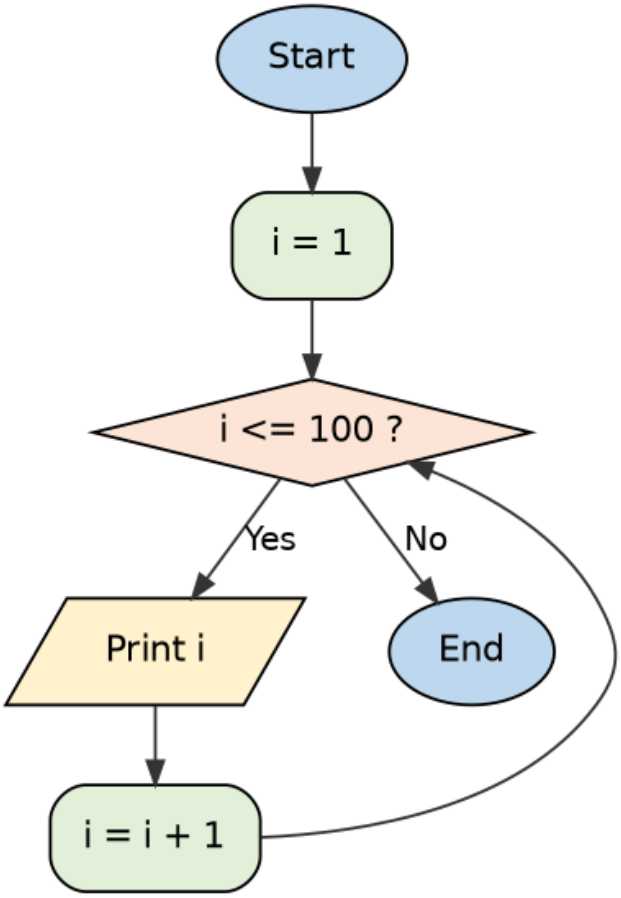
Output

```
Monday
```

Lab 14 – Loops

Question 14.1: Write a C program to print numbers from 1 to 100 using a for loop.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int i;

    for (i = 1; i <= 100; i++)
        printf("%d ", i);

    printf("\n");
    return 0;
}
```

Input

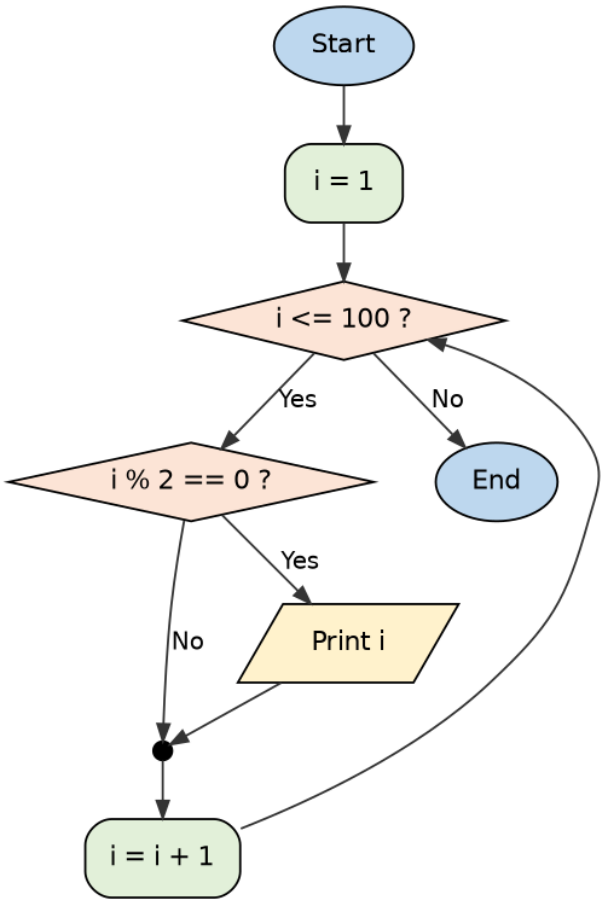
```
No input required.
```

Output

```
1 2 3 4 5 ... 98 99 100
```

Question 14.2: Write a C program to print even numbers from 1 to 100.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int i;

    for (i = 1; i <= 100; i++) {
        if (i % 2 == 0)
            printf("%d ", i);
    }
    printf("\n");
    return 0;
}
```

Input

```
No input required.
```

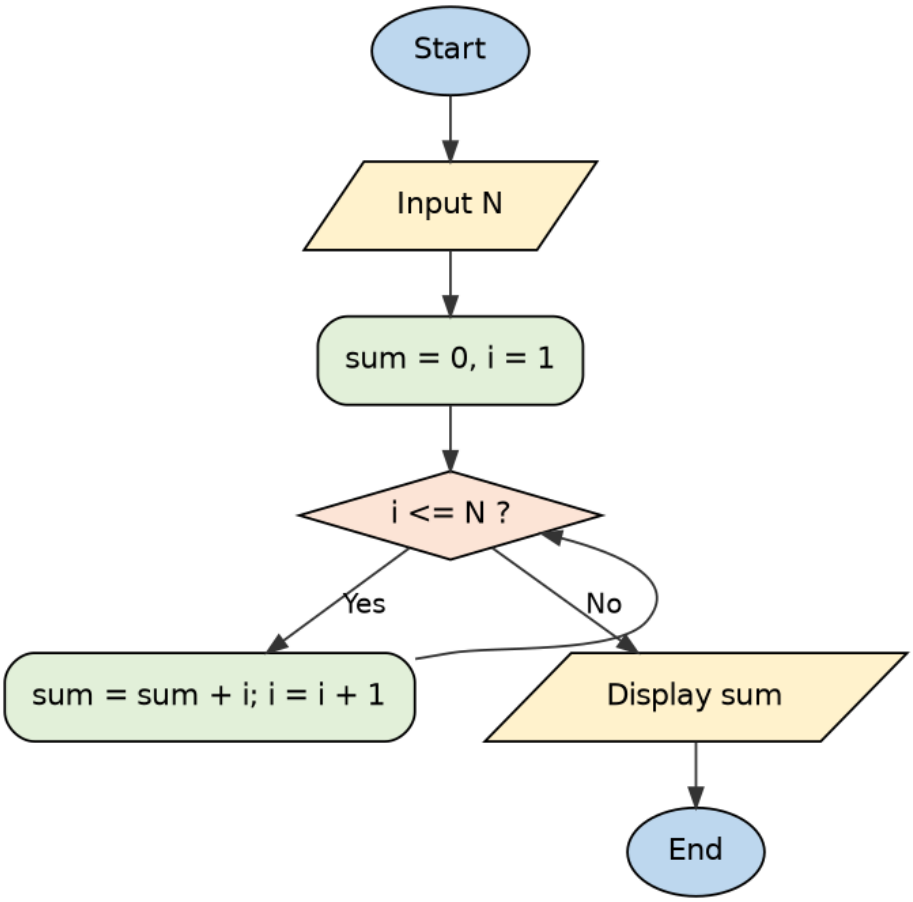
Output

```
2 4 6 8 10 ... 96 98 100
```

Lab 15 – for Loop Problems

Question 15.1: Write a C program to calculate the sum of the first N natural numbers.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n, i, sum = 0;

    printf("Enter N: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
        sum += i;

    printf("Sum of first %d natural numbers = %d\n", n, sum);
    return 0;
}
```

Input

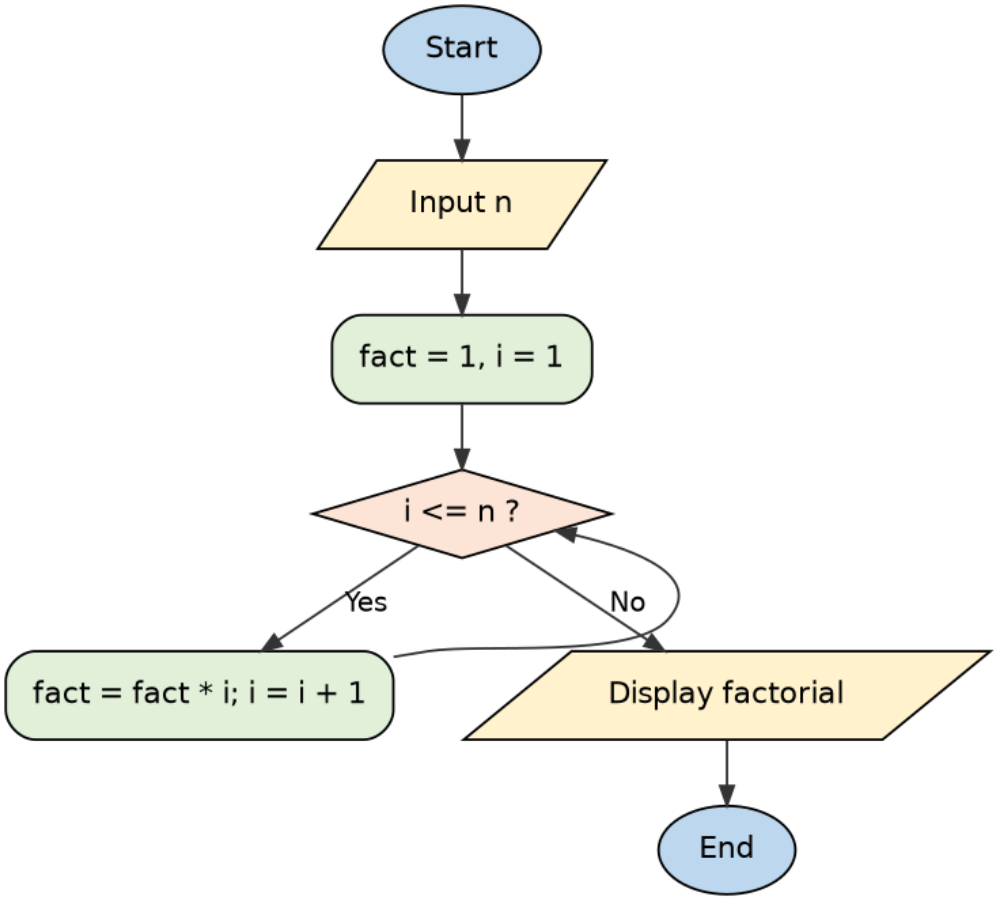
```
Enter N: 10
```

Output

```
Sum of first 10 natural numbers = 55
```

Question 15.2: Write a C program to calculate the factorial of a number.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n, i;
    unsigned long long fact = 1;

    printf("Enter a non-negative integer: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
        fact *= i;

    printf("Factorial of %d = %llu\n", n, fact);
    return 0;
}
```

Input

```
Enter a non-negative integer: 5
```

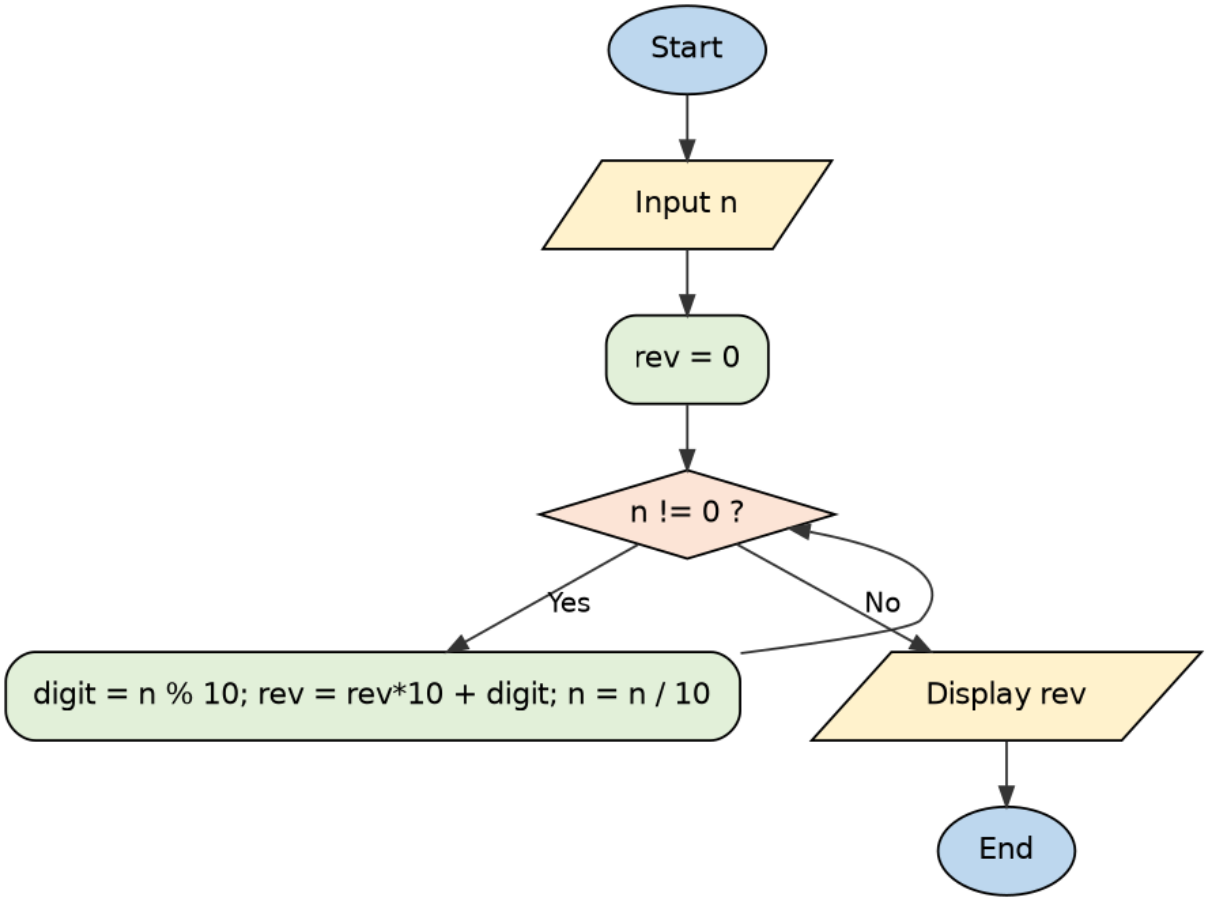
Output

```
Factorial of 5 = 120
```

Lab 16 – while and do-while

Question 16.1: Write a C program to reverse the digits of an integer using a while loop.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n, rev = 0, digit;

    printf("Enter an integer: ");
    scanf("%d", &n);

    while (n != 0) {
        digit = n % 10;
        rev = rev * 10 + digit;
        n /= 10;
    }

    printf("Reversed number = %d\n", rev);
    return 0;
}
```

Input

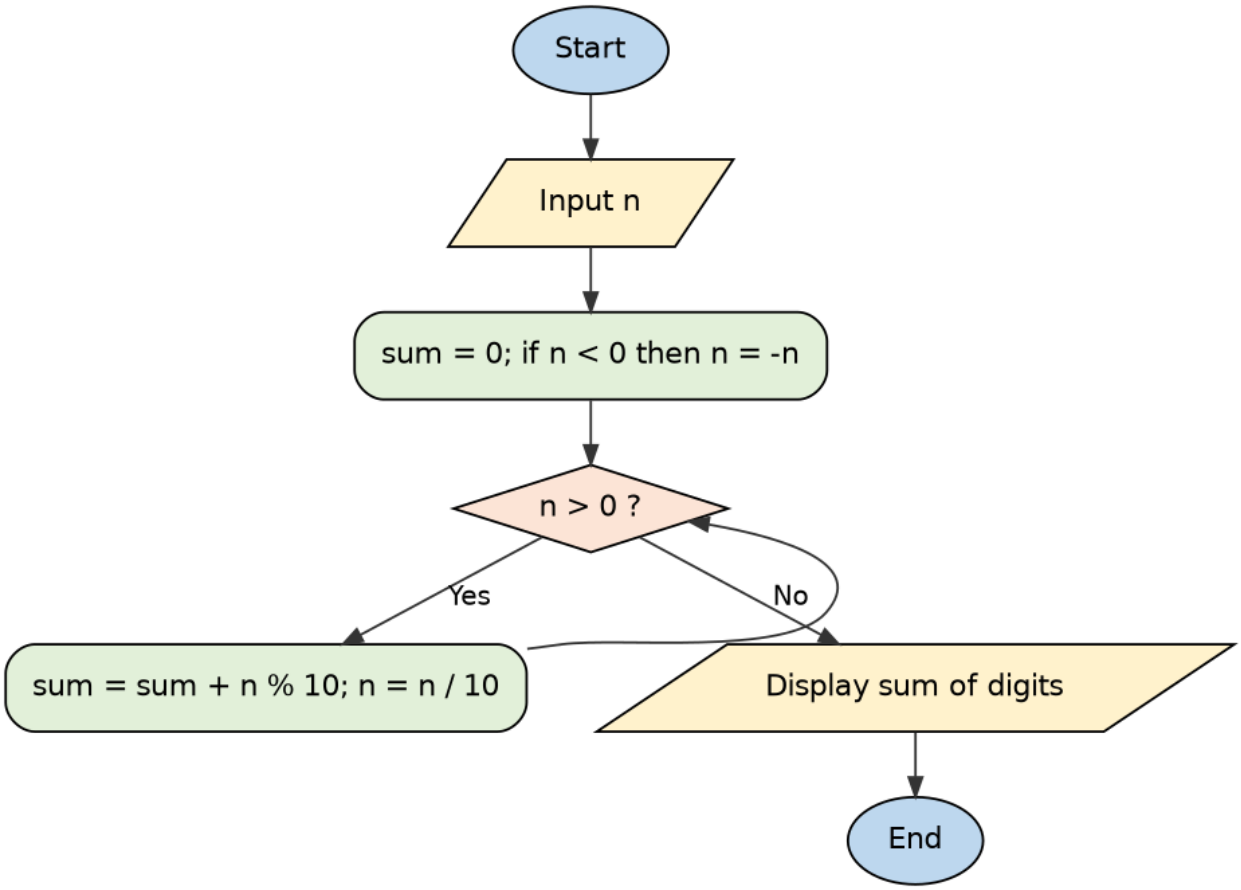
```
Enter an integer: 1234
```

Output

```
Reversed number = 4321
```

Question 16.2: Write a C program to find the sum of the digits of a number.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n, sum = 0;

    printf("Enter an integer: ");
    scanf("%d", &n);

    if (n < 0) n = -n;           /* handle negative input */

    while (n > 0) {
        sum += n % 10;
        n /= 10;
    }

    printf("Sum of digits = %d\n", sum);
    return 0;
}
```

Input

```
Enter an integer: 9875
```

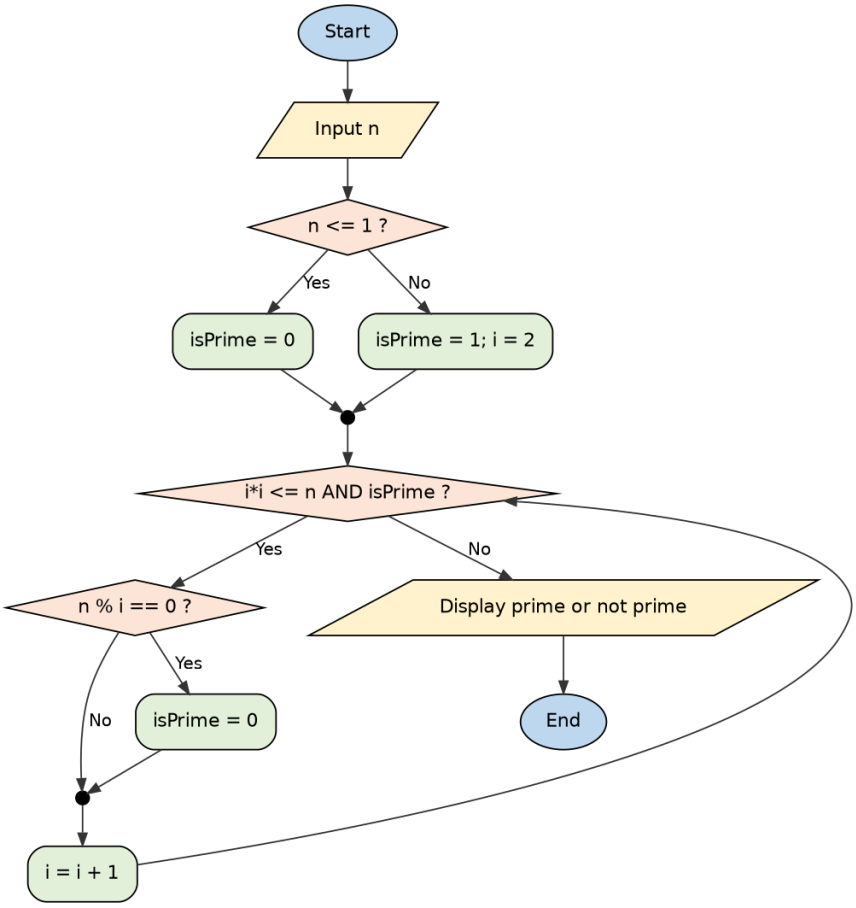
Output

```
Sum of digits = 29
```

Lab 17 – Number Problems

Question 17.1: Write a C program to check whether a number is a prime number.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n, i, isPrime = 1;

    printf("Enter a number: ");
    scanf("%d", &n);

    if (n <= 1)
        isPrime = 0;
    for (i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            isPrime = 0;
            break;
        }
    }

    if (isPrime)
        printf("%d is a prime number.\n", n);
    else
        printf("%d is not a prime number.\n", n);
    return 0;
}
```

Input

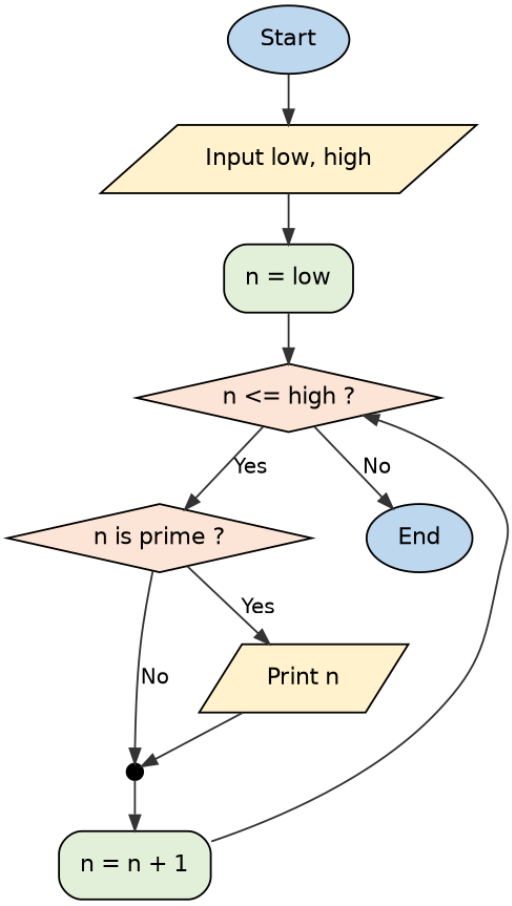
```
Enter a number: 29
```

Output

```
29 is a prime number.
```

Question 17.2: Write a C program to print all prime numbers between two given numbers.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int low, high, n, i, isPrime;

    printf("Enter lower and upper limits: ");
    scanf("%d %d", &low, &high);

    printf("Prime numbers between %d and %d:\n", low, high);
    for (n = low; n <= high; n++) {
        if (n <= 1)
            continue;
        isPrime = 1;
        for (i = 2; i * i <= n; i++) {
            if (n % i == 0) {
                isPrime = 0;
                break;
            }
        }
        if (isPrime)
            printf("%d ", n);
```

```
    }
    printf("\n");
    return 0;
}
```

Input

```
Enter lower and upper limits: 10 30
```

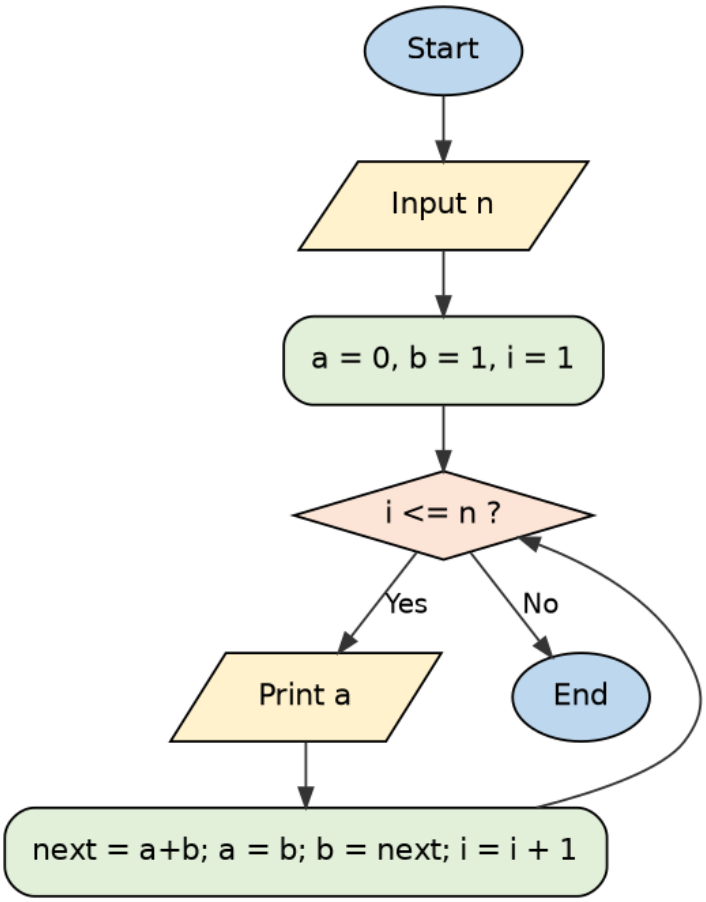
Output

```
Prime numbers between 10 and 30:
11 13 17 19 23 29
```

Lab 18 – Number Series

Question 18.1: Write a C program to generate the Fibonacci series up to N terms.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n, i;
    long long a = 0, b = 1, next;

    printf("Enter number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci series: ");
    for (i = 1; i <= n; i++) {
        printf("%lld ", a);
        next = a + b;
        a = b;
        b = next;
    }
    printf("\n");
    return 0;
}
```

Input

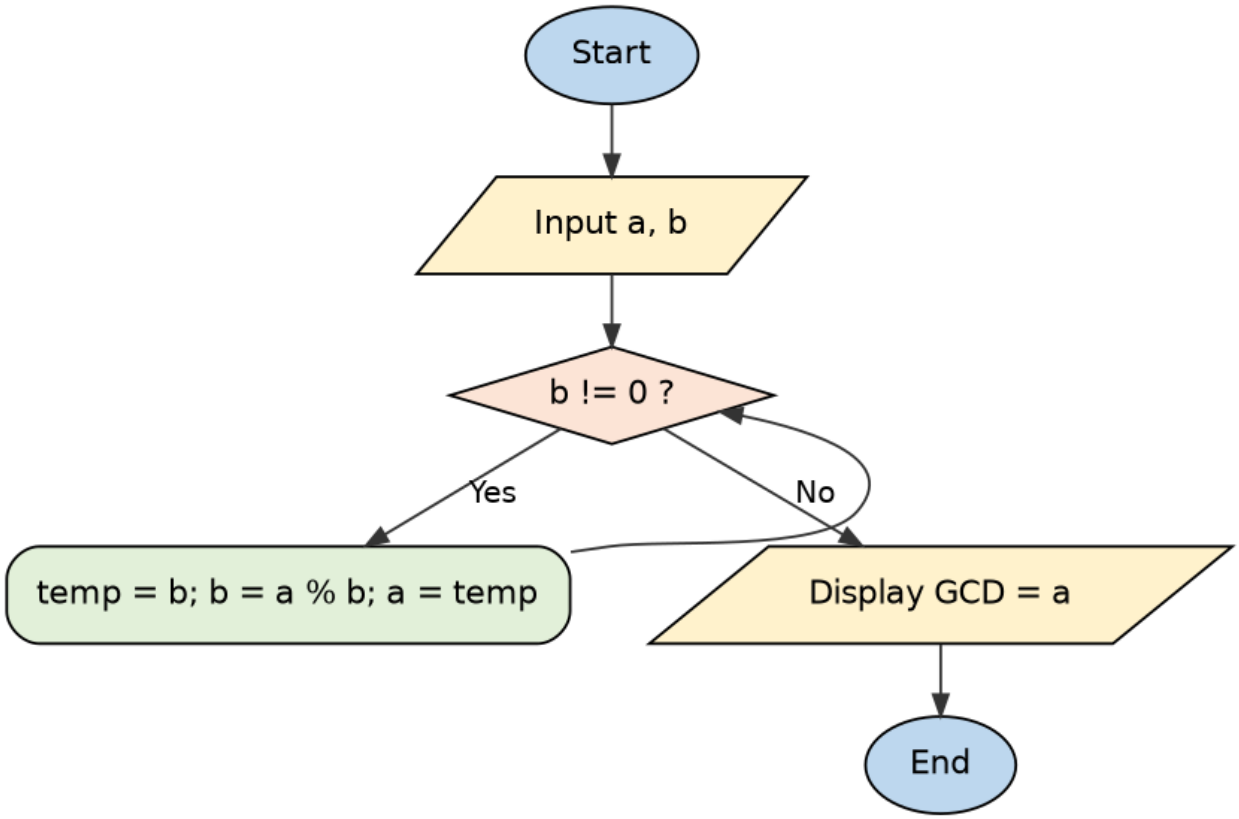
```
Enter number of terms: 10
```

Output

```
Fibonacci series: 0 1 1 2 3 5 8 13 21 34
```

Question 18.2: Write a C program to find the GCD of two numbers.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int a, b, temp;

    printf("Enter two positive integers: ");
    scanf("%d %d", &a, &b);

    while (b != 0) {             /* Euclidean algorithm */
        temp = b;
        b = a % b;
        a = temp;
    }

    printf("GCD = %d\n", a);
    return 0;
}
```

Input

```
Enter two positive integers: 48 18
```

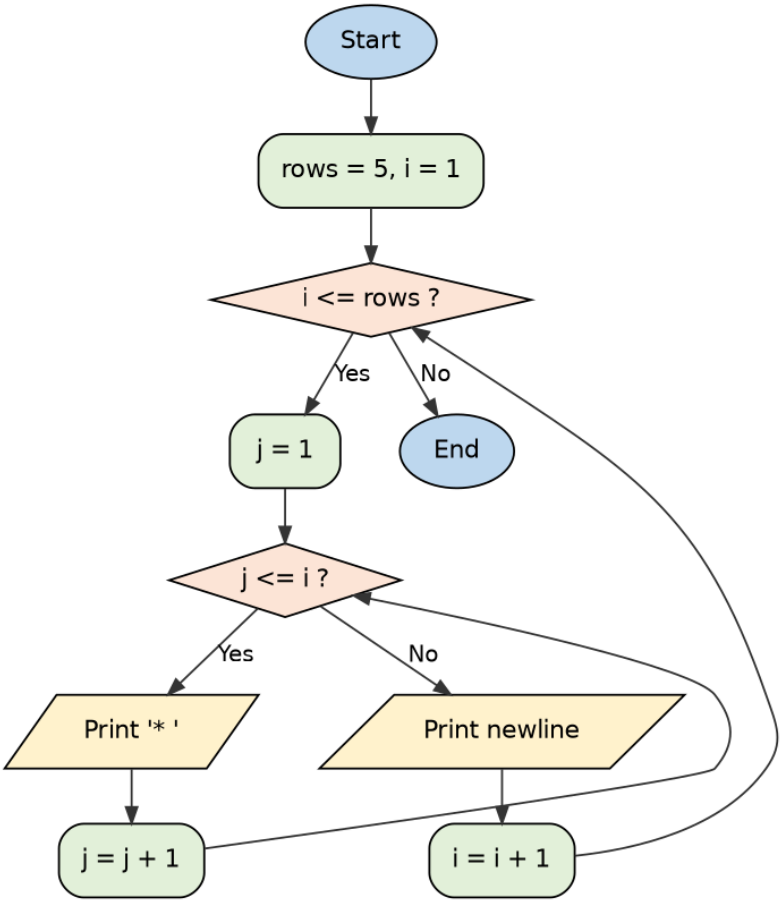
Output

```
GCD = 6
```

Lab 19 – Nested Loops

Question 19.1: Write a C program to print a right-angled star pattern.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int rows = 5, i, j;

    for (i = 1; i <= rows; i++) {
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
    return 0;
}
```

Input

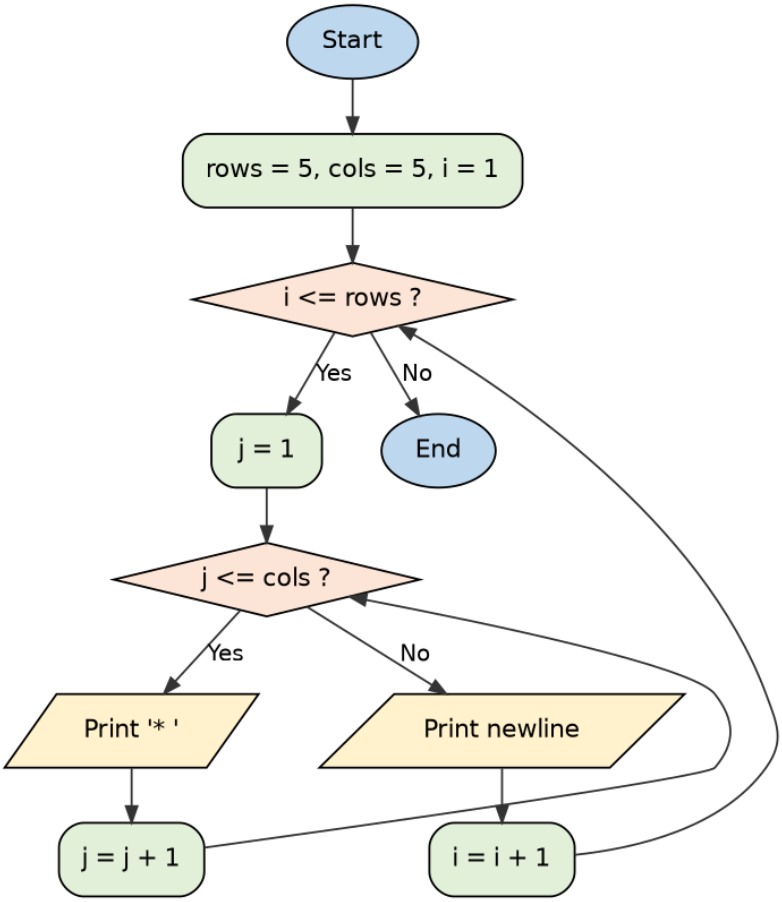
```
No input required.
```

Output

```
*
* *
* * *
* * * *
* * * * *
```

Question 19.2: Write a C program to print a square star pattern.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int rows = 5, cols = 5, i, j;

    for (i = 1; i <= rows; i++) {
        for (j = 1; j <= cols; j++)
            printf("* ");
        printf("\n");
    }
    return 0;
}
```

Input

```
No input required.
```

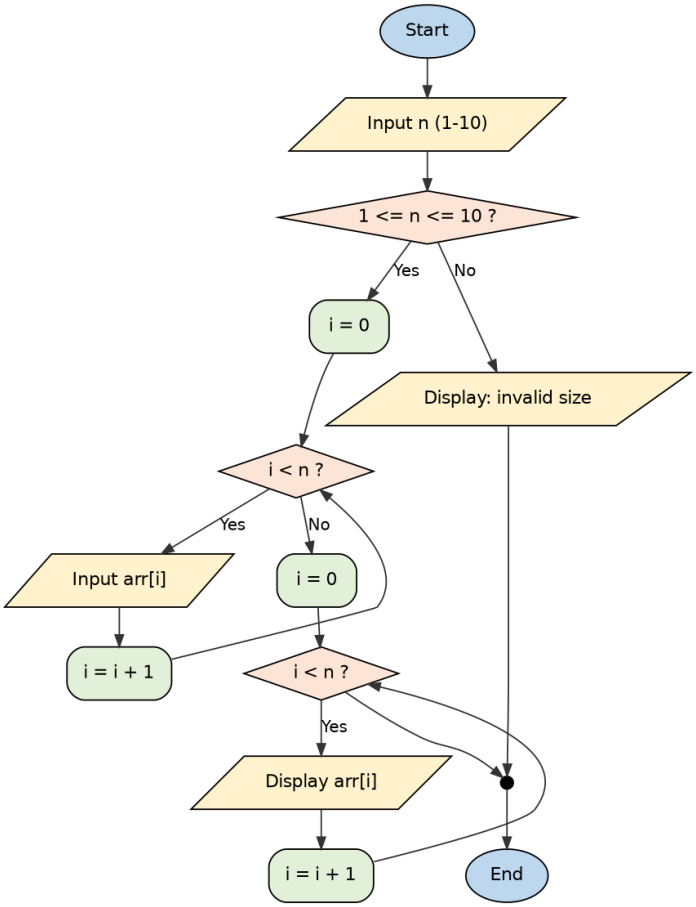
Output

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Lab 20 – Arrays

Question 20.1: Write a C program to input and display the elements of a one-dimensional array from 1 to 10 numbers.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n, i, arr[10];

    printf("How many elements (1-10)? ");
    scanf("%d", &n);

    if (n < 1 || n > 10) {
        printf("Invalid size.\n");
        return 1;
    }

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Array elements are:\n");
    for (i = 0; i < n; i++)
        printf("arr[%d] = %d\n", i, arr[i]);
    return 0;
}
```

Input

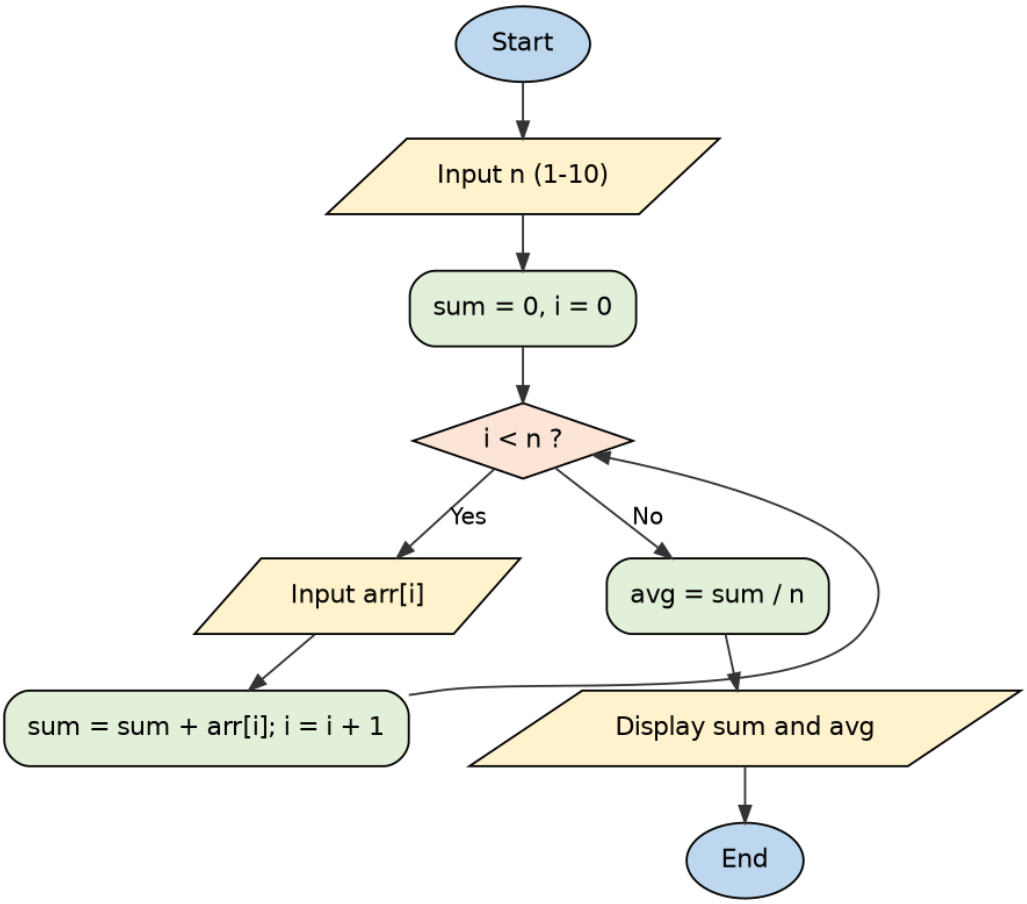
```
How many elements (1-10)? 4
Enter 4 elements: 7 3 9 1
```

Output

```
Array elements are:
arr[0] = 7
arr[1] = 3
arr[2] = 9
arr[3] = 1
```

Question 20.2: Write a C program to find the sum and average of array elements from user input.

Flowchart



Program

```
#include <stdio.h>

int main()
{
    int n, i, arr[10], sum = 0;
    float avg;

    printf("How many elements (1-10)? ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        sum += arr[i];
    }

    avg = (float)sum / n;
```

```
    printf("Sum     = %d\n", sum);
    printf("Average = %.2f\n", avg);
    return 0;
}
```

Input

```
How many elements (1-10)? 5
Enter 5 elements: 2 4 6 8 10
```

Output

```
Sum     = 30
Average = 6.00
```